



CHAPTER FOUR

WATER APPLICATION METHODS

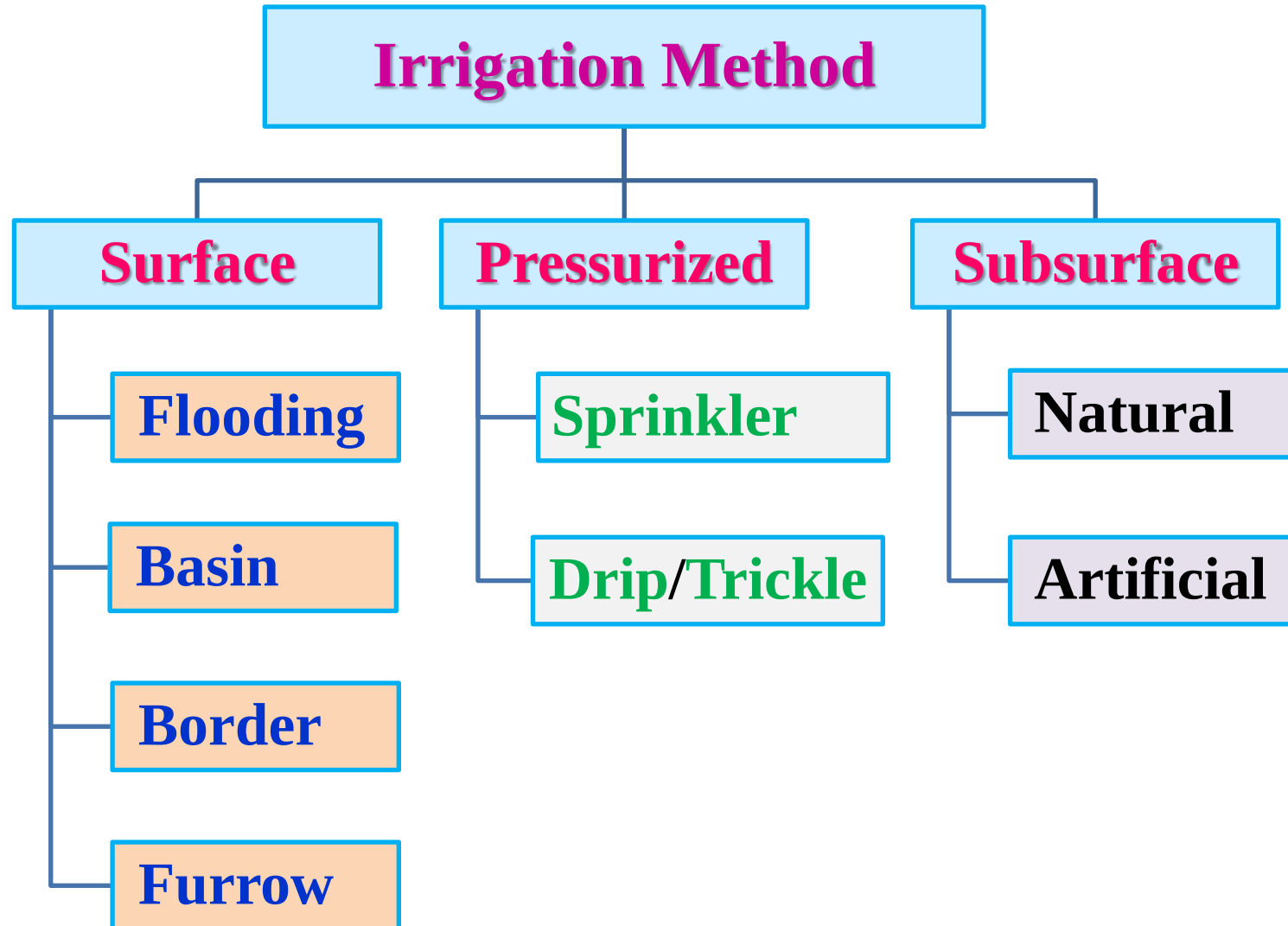
Chapter 4

Water Application Methods

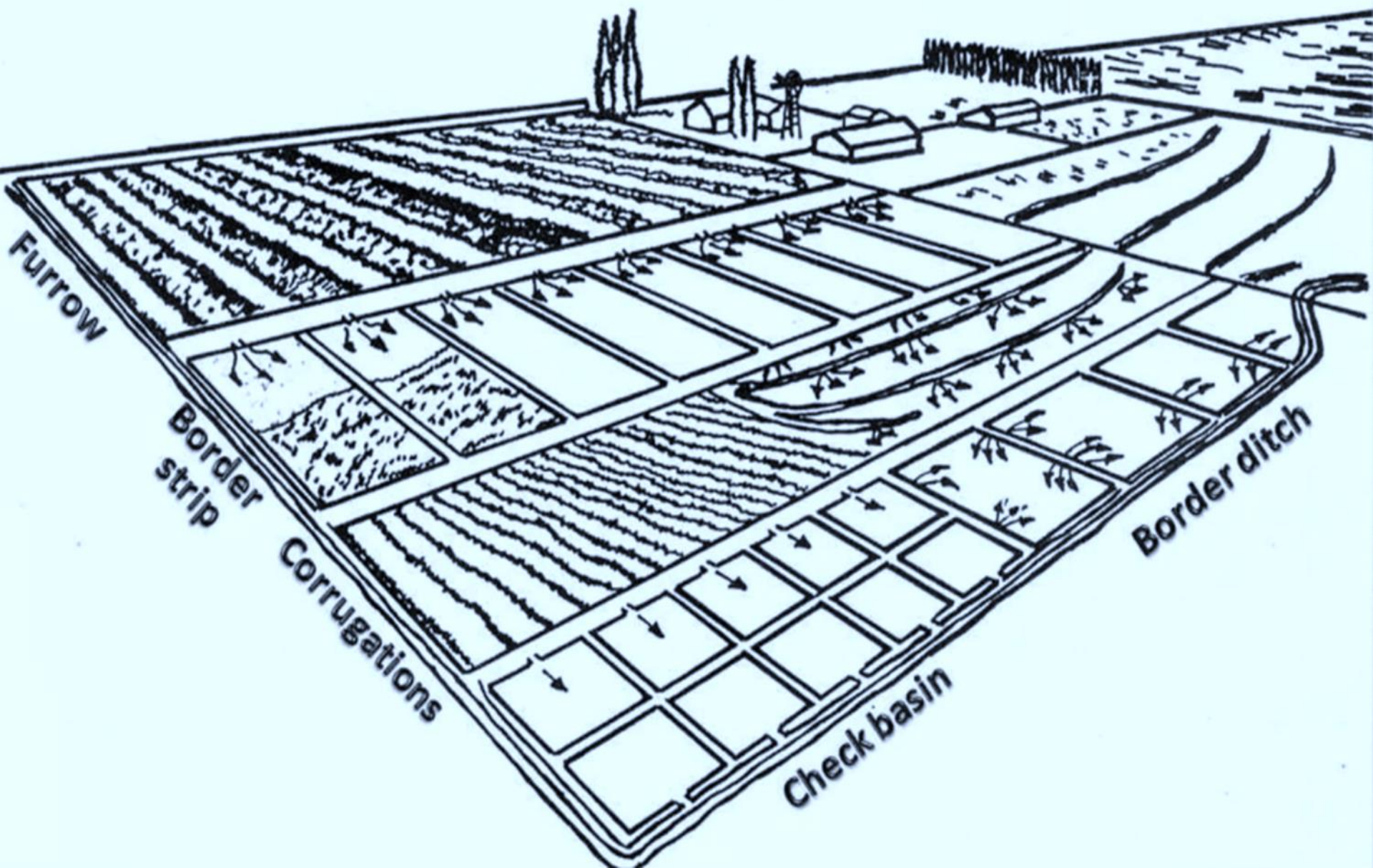


Chapter 4

Water Application Methods



Surface Irrigation Methods



Surface Irrigation Methods

- Is a broad class of irrigation methods, in which water is applied/distributed over the field by overland flow (gravity flow).
- A flow is introduced at one edge of the field and covers the field gradually.
- Surface irrigation is entirely practiced where water is abundant.
- The rate of coverage (advance) is dependent almost entirely on the differences between the discharge onto the field and the accumulating infiltration into the soil.
- Secondary factors include field slope, surface roughness, and the geometry or shape of the flow cross-section.

Surface Irrigation . . .

Components of a surface irrigation system:

- Physical system of surface irrigation consist of:
 - ⇒ The water source
 - ⇒ The water supply subsystem
 - ⇒ The water delivery subsystem
 - ⇒ The water use subsystem
 - ⇒ The water removal subsystem

The water source

- The source of water can be surface water or groundwater.
- Water can be abstracted from a river, lake, reservoir, borehole, well, spring, etc.

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Components of . . .

Water supply subsystem

- Diverts water from the source to conveyance /delivery system (e.g weir, barrage . . .)

Water delivery sub-system

- Conveys water from the source to field through canals, distributaries, & channels at regulated rate

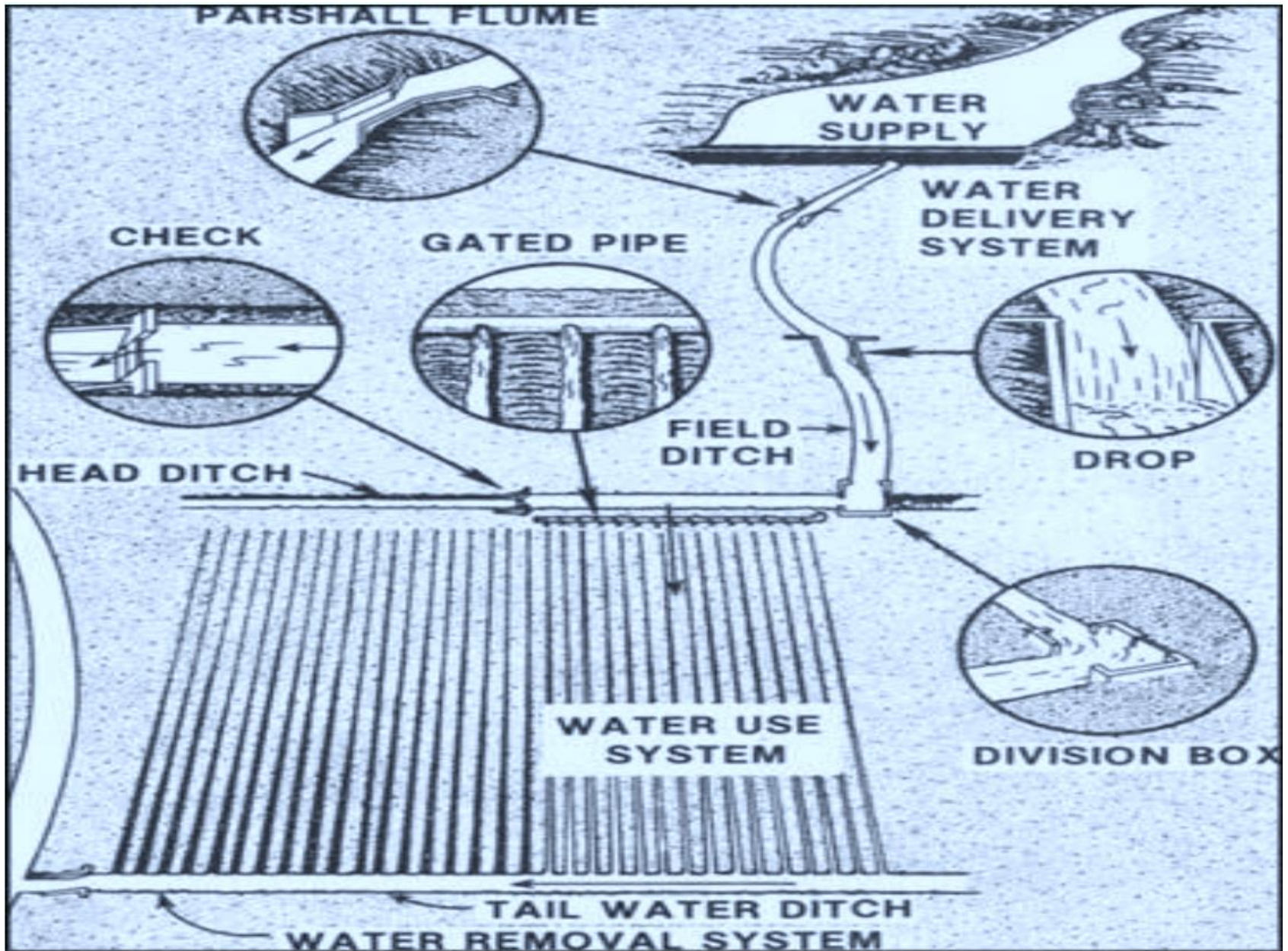
Water use sub- system

- Receives water from the water delivery & application sub-systems

Water removal sub –system

- Used for removal and disposal of excess surface and sub- surface water to improve aeration & salinity levels

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General characteristics of Surface Irrigation:

- Old-age method
- the most extensively used method worldwide
- water application is directly on the soil surface
- Requires relatively minimal capital investment
- doesn't require complicated and expensive equipment
- Energy costs are substantially lower
- Labor requirements is relatively high
- Relatively inefficient method
- Limited to land already having small and even slopes
- entirely practiced where water is abundant.
- More affected by water logging and salinity problems
- Land leveling costs are high

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Advantages and disadvantages of surface irrigation

Advantages

- It is more acceptable to agriculturalists or requires less technical skills to operate & maintain.
- Energy requirements come from gravity.
- Less affected by climatic and water quality characteristics.
- If the topography is not too undulating, these costs are not great.
- Leaching is easier and cheaper if it is needed.
- Use large flows available for short periods, and so allows canalized water supply to be easily shared by several farmers.

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Advantages and disadvantages of surface irrigation . . .

Disadvantages

- ❖ Difficult to get even distribution of water on light permeable soils which have high infiltration rates
- ❖ Greater loss of water by evaporation, surface runoff, deep percolation.
- ❖ Low efficiency due to imperfect control over the water flow.
- ❖ Not suitable for crops which need frequent light watering (shallow rooted, drought vulnerable crops).
- ❖ Usually requires smooth land. Costly and time consuming preparation of land/land grading.
- ❖ Field layout for surface irrigation may restrict mechanization unless special measures are adopted.
- ❖ Increase water born disease.

Types of Surface Irrigation

- Water is applied to the field in either the **controlled** or **uncontrolled** manner.
 - Surface Irrigation Methods can be further classified as:
 - Wild Flooding Irrigation: **Uncontrolled**
 - Furrow Irrigation
 - Basin Irrigation
 - Border Irrigation
- Water is applied from the head ditch and guided by corrugations, furrows, borders, or ridges.

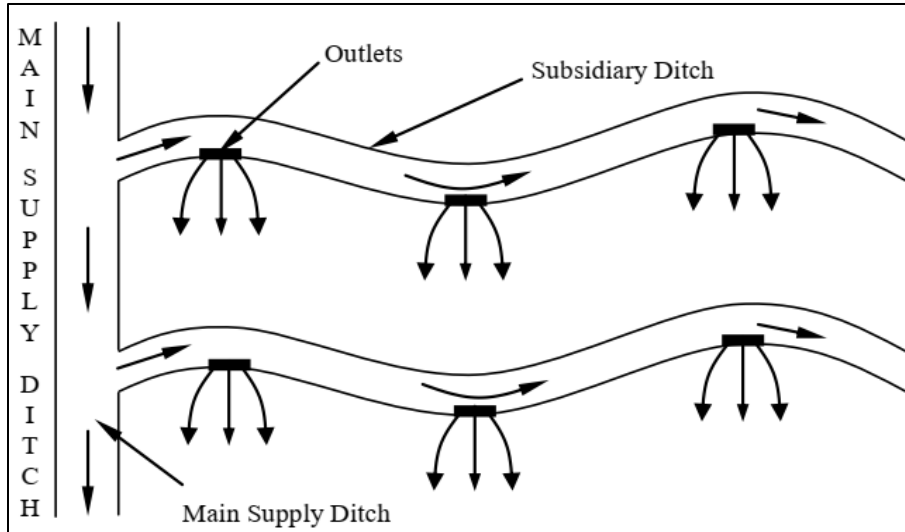
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❖ Wild Flooding

- In this method, **ditches are excavated in the field** either on the contour or up and down the slope.
- Water from these ditches applied/flows across the field, since the **movement of water is not restricted or controlled** by means of levees or land shaping, it is called **wild flooding**.
- Recommended **only when irrigation water is abundant (no water scarcity)** and where **land levelling is not followed**.
- It involves low initial cost of land preparation, extra labor cost in the application of water.
- **Application efficiency is low** due to deep percolation or excess runoff
- Most **suitable to close growing crops, pastures**, etc. or the crop values do not justify adoption of better method
- May be **used on rolling land of irregular topography**, where borders, basins and furrows are not feasible.

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❖ Wild Flooding . . .



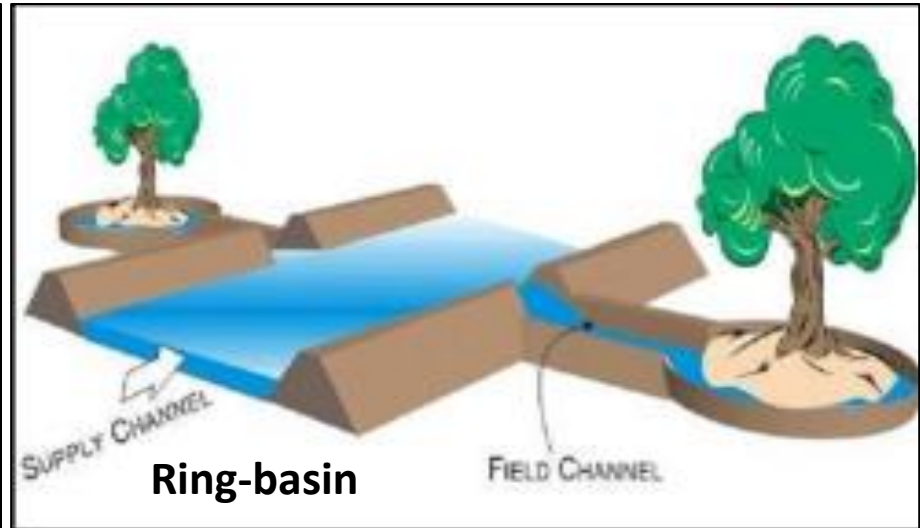
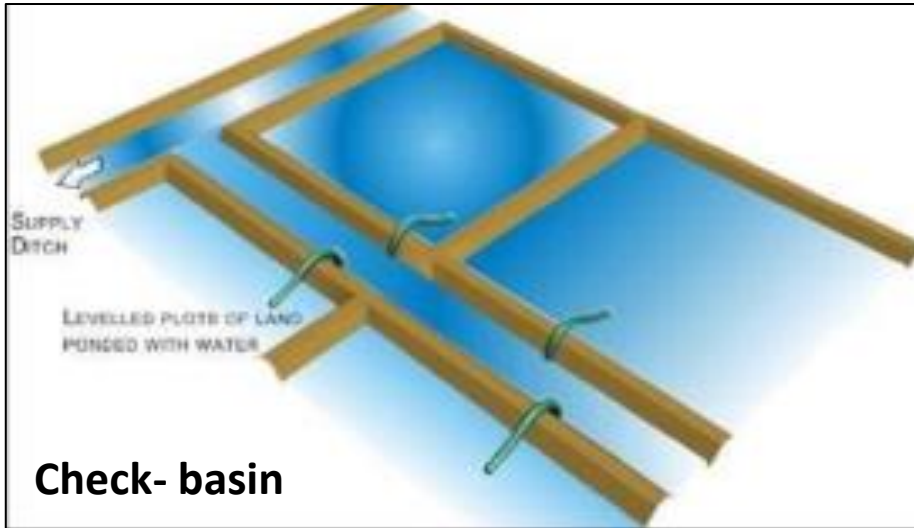
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❖ Basin Irrigation

- Involves dividing the field into a number of small strips of land called **basins**.
- Basins are generally a level or flat areas of land, surrounded by low ridges (bunds, dykes, levees) to guide water, in w/c water can be ponded until it infiltrates into the soil.
- The dykes/levees/bunds prevent runoff from basin, and also undirected flow onto the basin.
- Water is filled quickly in the basin and is allowed to percolate into the soil.
- Suited to different crops, such as, rice, cotton, groundnuts, trees, orchards (fruit trees) etc
- And to fine textured soils of moderate to low intake rate having smooth, gentle and uniform land slopes.

Cont'd . . .

❖ Basin Irrigation . . .



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❖ Basin Irrigation . . .

- Generally, basin irrigation is recommended for:
 - Salt affected area
 - For closely spaced crop and sensitive to flooding
 - Soil with low infiltration rate
 - Slope less than 0.1% flat lands.

Size of Basins:

- It depends on soil type, stream size, irrigation depth, field size, etc.

Generally,

- Basins are much larger on clay soils than on sandy
- Basins can be larger when a large stream size is used.
- Basins can be larger when a larger irrigation depth is used,

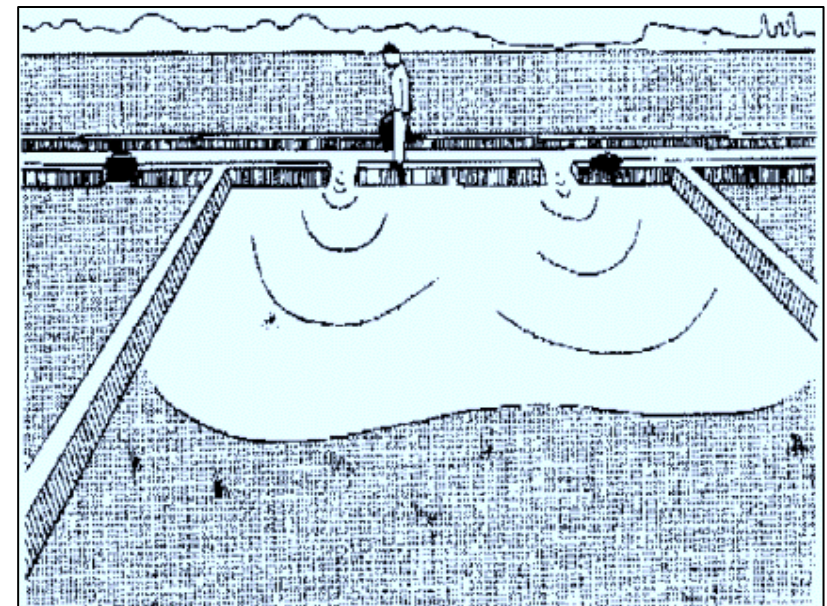
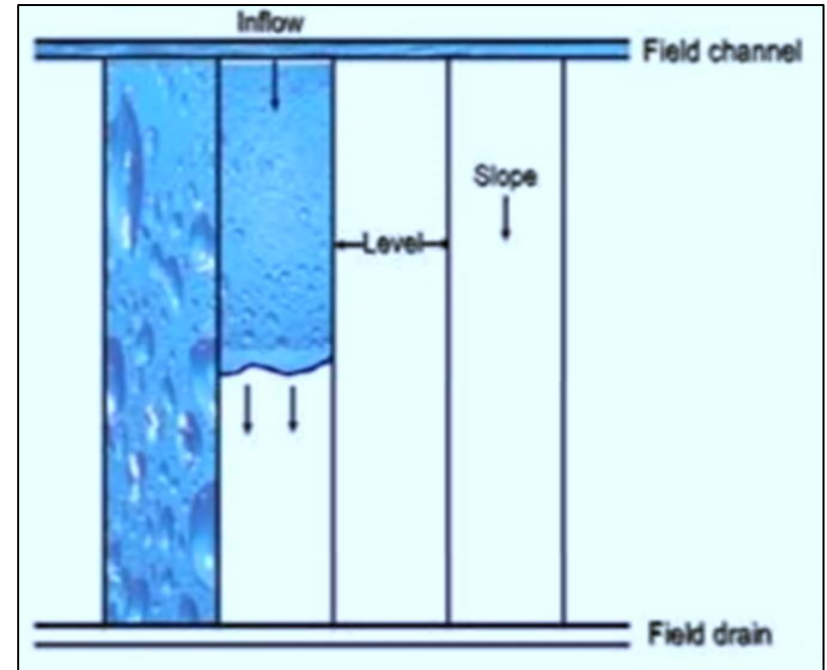
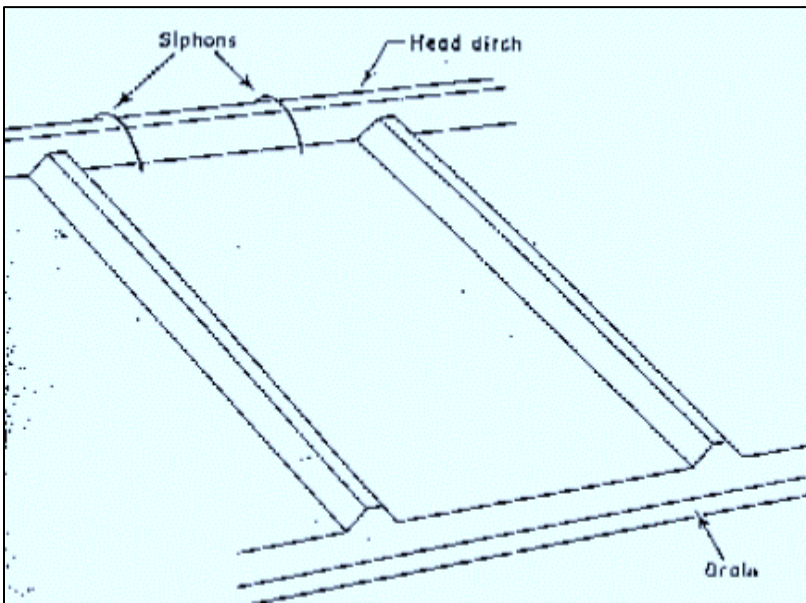
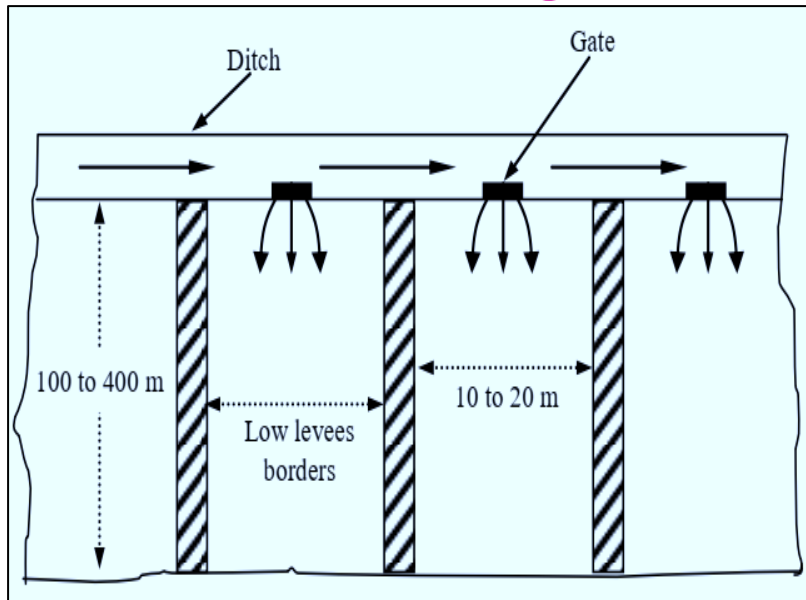
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❖ Border Strip Irrigation

- Here, the land is divided into strips (border) by parallel small earth bunds or dykes called **border ridges/ border strips**.
- **Each strip is irrigated** by introducing water at u/s from the supply ditch and it progressively covers the entire strip.
- The water flows slowly towards the lower end, and infiltrates into the soil as it advances.
- When the advancing water reaches the lower end of the strip, the supply of water to the strip is turned off.
- When the water is shut off, it recedes from the upper end to the lower end.
- The **longitudinal/lengthwise slope must be uniform** (0.2 ~ 0.4%), and the **transverse slope must be level or zero or negligible** (< 0.03%).

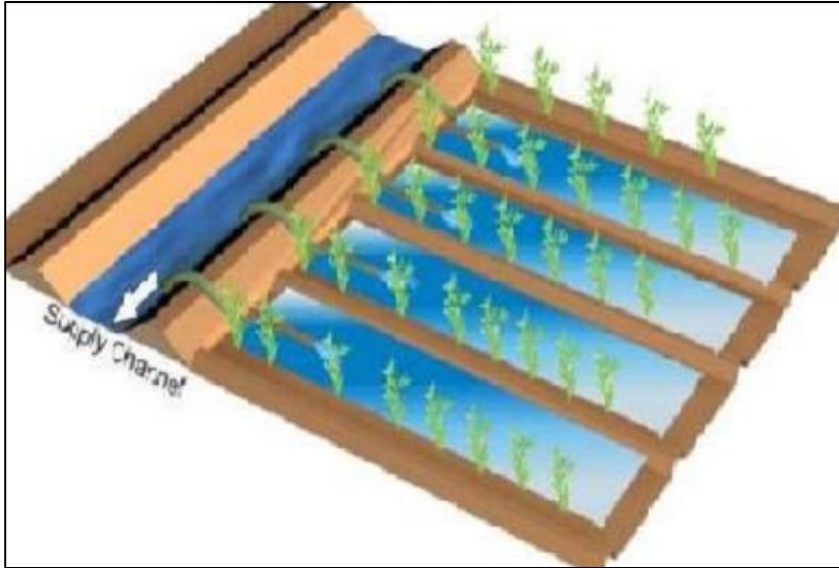
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❖ Border flooding



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❖ Border flooding



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❖ **Border flooding . . .**

- Suited to soils between extremely high and very low infiltration rates.
 - Coarse textured soils with high infiltration rates will require high discharge rate (large supply ditch), in order to spread water over the entire strip rapidly. and to avoid excessive losses due to deep percolation at the upper reaches.
 - Fine textured soils with low infiltration rates, require smaller ditches (low discharge rate) to avoid excessive losses due to run off at the lower reaches.
- Suitable for high value crops or crops that can withstand flooding for a short time (close growing crops) e.g wheat, barley, etc. and used for some row crops like cotton.

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❖ Border flooding . . .

Size and shape of Borders:

- › Borders are often **long** and **narrow**.
- › They are usually **rectangular in shape** varying in size from 100 - 800 m long and 3 - 30 m in width.
- › The size and shape of the border strips **depends** on **types of soil, stream size, irrigation depth, slopes of the land, field size**.
- › Generally, Borders;
 - are much longer on clay soils than on sandy soils.
 - are longer when a larger unit stream size is available.
 - are longer when a larger irrigation depth is applied.
 - must be shorter on steeper sloping land to prevent erosion.
 - can be wider when larger stream sizes are available.

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❖ Border flooding . . .

Soil type	Border Slope (%)	Unit flow per metre width (l/sec)	Border Width (m)	Border Length (m)
SAND	0.2-0.4	10-15	12-30	60-90
Infiltration rate greater than 25 mm/h	0.4-0.6	8-10	9-12	60-90
	0.6-1.0	5-8	6-9	75
LOAM	0.2-0.4	5-7	12-30	90-250
Infiltration rate of 10 to 25 mm/h	0.4-0.6	4-6	6-12	90-180
	0.6-1.0	2-4	6	90
CLAY	0.2-0.4	3-4	12-30	180-300
Infiltration rate less than 10 mm/h	0.4-0.6	2-3	6-12	90-180
	0.6-1.0	1-2	6	90

Cont'd . . .

❖ Border flooding . . .

Advantage

- › Large water streams can be used safely
- › Provides uniform wetting of soil profile
- › Low labor requirement and time,
- › Low maintenance cost

Disadvantage

- › Proper leveling and high initial cost
- › Requires relatively large water stream (large supply of water) for quick advance of water to minimize deep percolation losses at the upper end of the border strip.
- › Wastage of water by deep percolation in coarse textured soils

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❖ Furrow Irrigation

- Is the most widely used method for irrigating row crops.
- A furrow consists of a narrow ditch or channel between the rows of crops.
- Water is no longer flows over the entire soil surface but is applied in small stream (channels or furrows) between the rows of crops, grown on ridges.
- It is then gradually absorbed into the bottom and sides of the furrow to wet the soil.

Water infiltrates through the wetted perimeter and spreads vertically and horizontally to refill the soil reservoir.

- The furrows must all have equal dimensions, in order to guarantee that the water is distributed evenly.

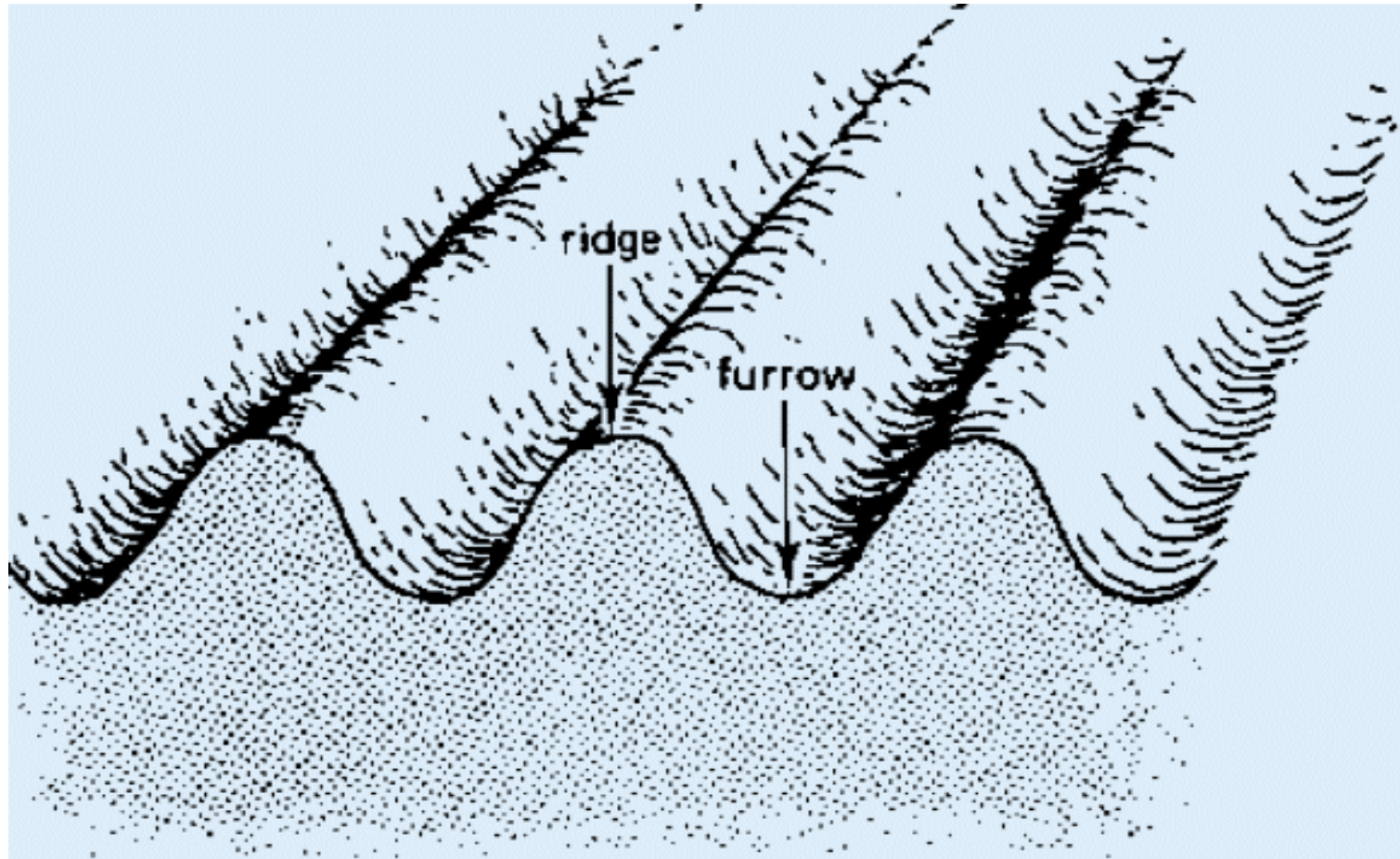
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❖ Furrow Irrigation . . .



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❖ Furrow Irrigation . . .



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❖ Furrow Irrigation . . .

- Like flood irrigation, furrow irrigation is rather **cheap in areas where water is inexpensive**.
- In furrow irrigation, only a part of (only half to one-fifth of) the irrigated land surface (the furrow) is wetted thus minimizing evaporation losses.
- Furrow irrigation is **adapted for** irrigating on **various slopes except on steep ones** because of erosion and bank overflow.
- but is **not suitable for very light soils having high infiltration capacity**, as the water is wasted on the upper end of the furrow due to deep percolation.
- Furrow irrigation is **adaptable to all row crops**

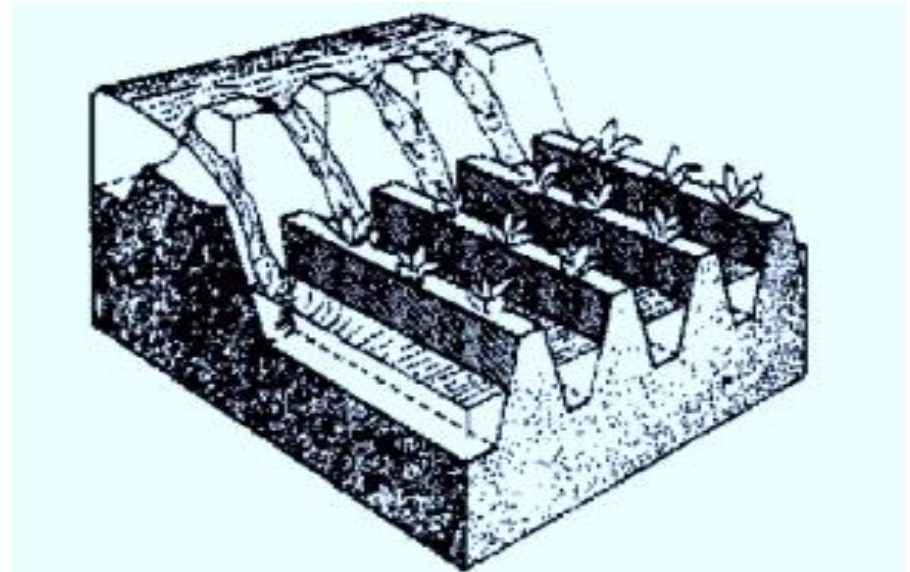
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❖ Furrow Irrigation

- There are different ways of applying water to the furrow from the head ditch.
 - **Using siphons:** siphons are used to divert water from the head ditch to the furrows.
 - **by cutting the ridge or levee separating the head ditch and the furrows and by let it flow direct gravity.**
 - **Using spill (gated short pipes) buried in the ridges.**



By Siphons



By cutting the ridge

Cont'd . . .

❖ Furrow Irrigation

- The size and shape of the furrow depend on the soil type, crop type, cultivation practice (equipment used for the formation of furrow, cultivation and harvesting), stream size, irrigation depth, field size and slope.

Furrow length:

- It depends on soil type, stream size, irrigation depth, field size and shape, slope, farming practice.

But generally, Furrows,

- can usually be longer on clay soils than on sandy soils.
- can be longer when a larger stream size is used for irrigation.
- can be longer when a larger irrigation depth is applied.
- are usually shorter on steeper sloping land to prevent erosion.

❖ **Furrow Irrigation**

Furrow Spacing: depends on

- Water movement in the soil
 - E.g. close furrows on sandy soils
- Crop
 - E.g. double rows for onions
- Cultivation Practice
 - E.g. equipment's used for cultivation & harvesting

Furrow Slope:

- Ideally furrows should have a uniform slope.
- A minimum slope of 0.05% is needed to ensure that water will flow down the furrow and any excess water can be drained.
- The maximum slope depends on the risk of soil erosion.

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❖ Furrow Irrigation

Crops:

- Most row crops such as vegetables, cotton, sugar beet and potatoes
- The crops are usually planted on raised bed or ridges

Irrigating furrows:

- When irrigating furrows, water is supplied to each furrow from the farm channel using siphons or spills
- Several furrows can be irrigated at the same time depending up on the discharge available in the farm channel
- The stream size must be large enough for the flow to advance quickly down the furrow.
- Runoff can be a major problem in furrow irrigation.

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❖ Furrow Irrigation

Advantage

- › Low evaporation loss;
- › Furrow serves as drainage ways for surface runoff of heavy rainfall areas;
- › Suitable for crops like maize that injured by contact with water.

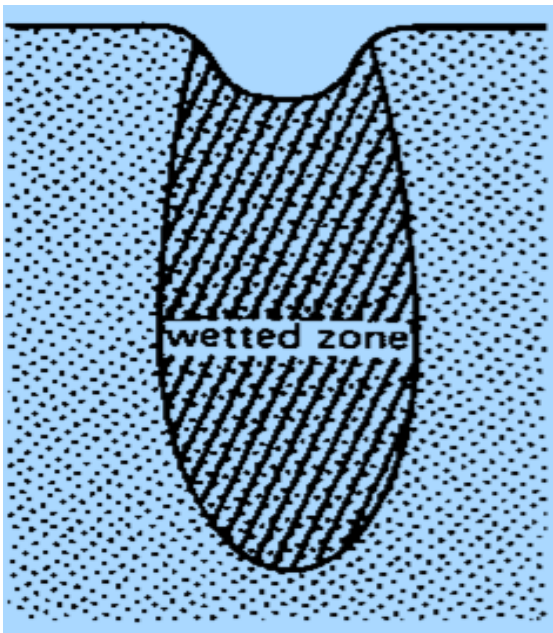
Disadvantage

- › Not recommended for soil having high infiltration capacity;
- › An accumulation of salinity between furrows;
- › An increased level of tail water losses;
- › Difficulty of moving farm equipment across the furrows.

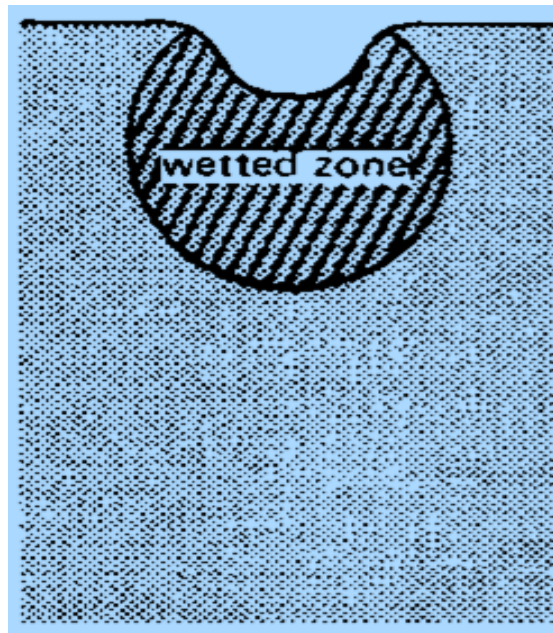
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❖ Furrow Irrigation

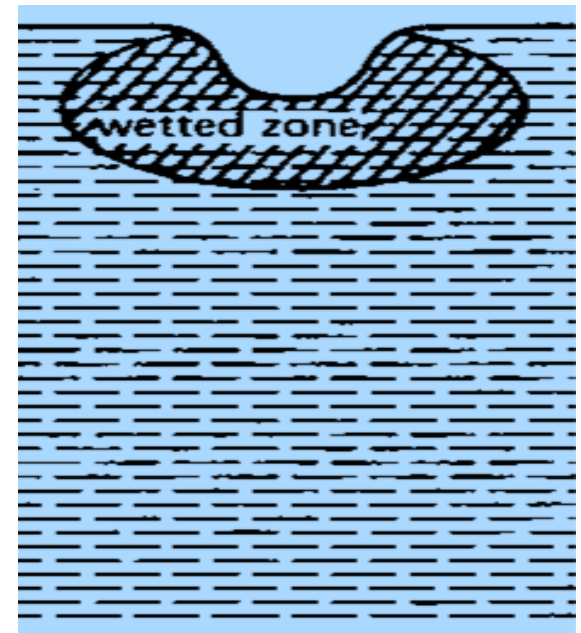
Wetted zone for different soil type



Sand soil



loam soil



Clay soil

Criteria for selection of Surface irrigation methods

- Soil type
- Type of crop to be irrigated,
- Required depth of application,
- Field shape,
- Previous experiences with irrigation,
- Labor availability.
- Slope of topography

Criteria for selection of Surface irrigation methods

Item	Basin	Border	Furrow
Main slope	Usually zero slope or nearly zero attained	Up to 2-5 % (min. 0.05%)	< 1 % (min. 0.05%)
Soils	medium to fine textured soils	moderately low to moderately high intake rate soils	best suited to soils with moderate to low intake rates.
Infiltration	one dimensional and vertically downward	same as basin	two dimensional
Field size	all size	Large	Large
Geometry shape of farm	all shape	Regular	Regular
Sediment load	not problematic	not problematic	not problematic
Biological quality	not problematic	not problematic	not problematic
Salinity	not problematic	slightly problematic	problem if very high salt

Criteria for selection of Surface irrigation methods

Item	Basin	Border	Furrow
Crops variety	all crops but best for ponded water crops	best for close growing crops	best for row crops
Farming machinery	difficult to use	easy to apply	adapted to mechanized farming
Labor input	least labor compared to other surface irrigation systems	High	High
Application efficiency	65 – 80 %	60 - 70 %	60 - 70%
Level of technology requirements	simplest	Highest	lower than border
Initial cost	higher	Lower	low to medium

Phase of surface irrigation

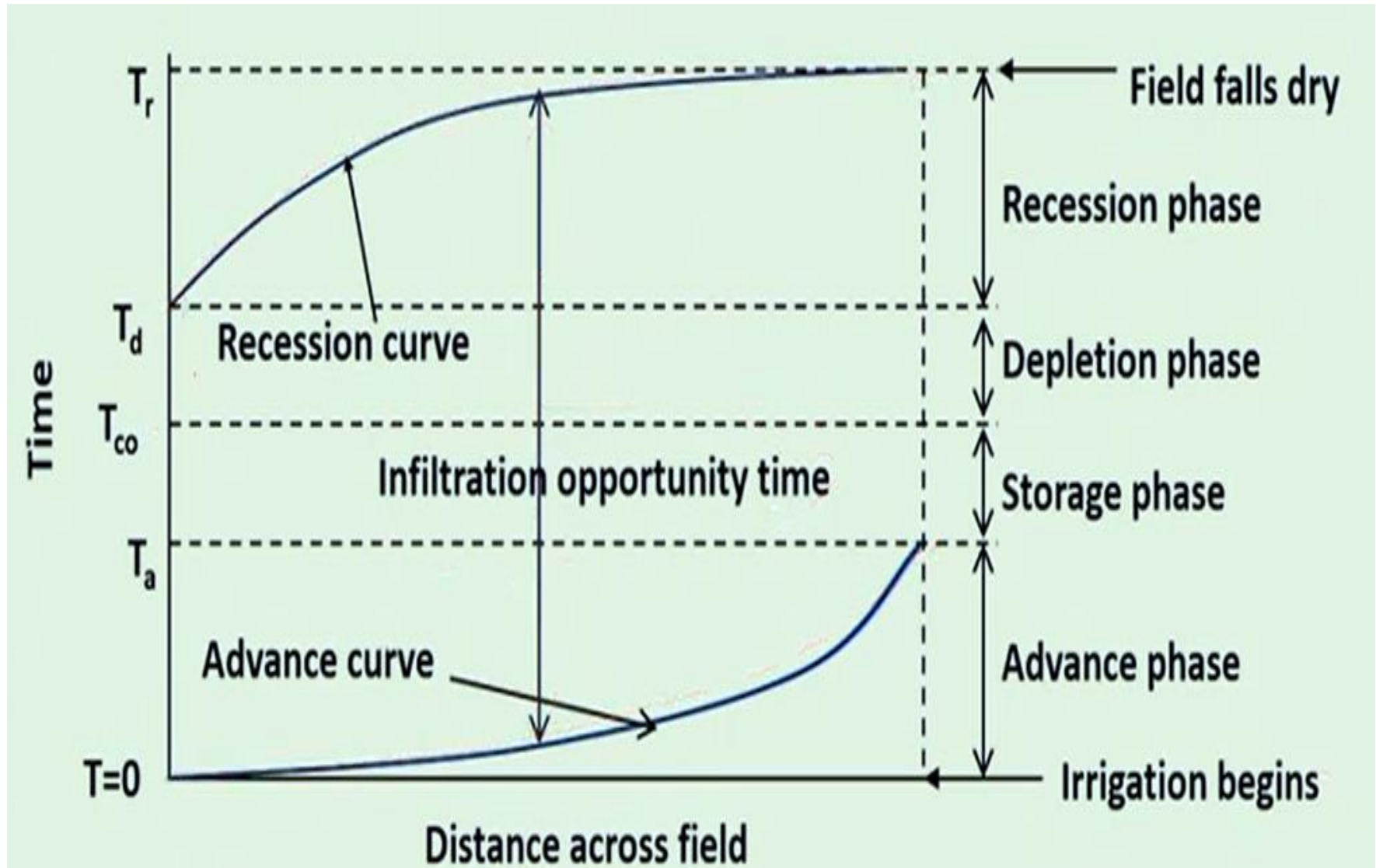
- ✧ **Advance phase:** Time during which water advances from upper end to the lower end of the field.
- ✧ **Storage or Ponding phase:** Time elapse between the end of advance (arrival of water at the tail end) and inflow cut-off (the stopping of inflow at the top/head end).
- ✧ **Depletion Phase:** Time between inflow cut-off and the beginning of recession (the appearance of the first bare soil) at upper end of the field.

Also known as the recession lag time or known as vertical recession.

Depth of flow at upstream end decreases from normal depth to zero.

- ✧ **Recession Phase:** Time between the beginning of recession at the upper field end and disappearance of water from the field surface

Phase of surface irrigation . . .



Phase of surface irrigation . . .

❖ Advance phase:

- Begins when water is turned into the field
- Ends when water reaches the d/s end of the field

❖ Storage or Ponding phase:

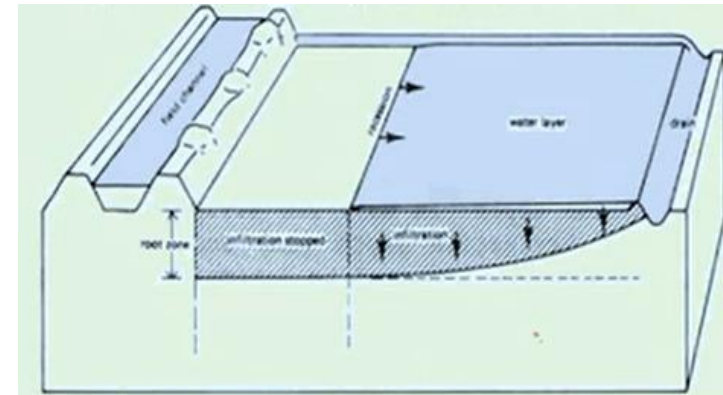
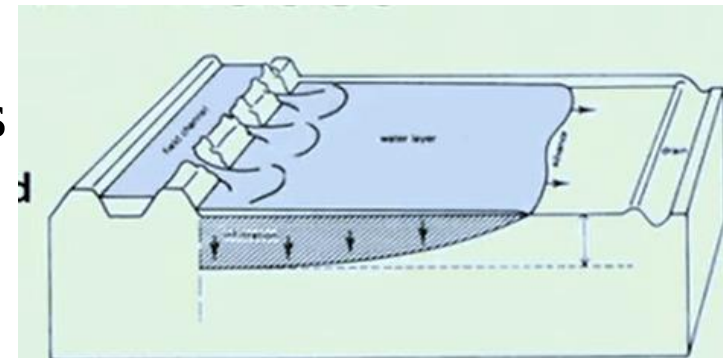
- Begin when the advance phase ends
- This phase ends when inflow ends

❖ Depletion Phase:

- Begins when storage phase ends (when inflow ends)
- Ends when depth of flow at the headend become zero (recess)

❖ Recession Phase:

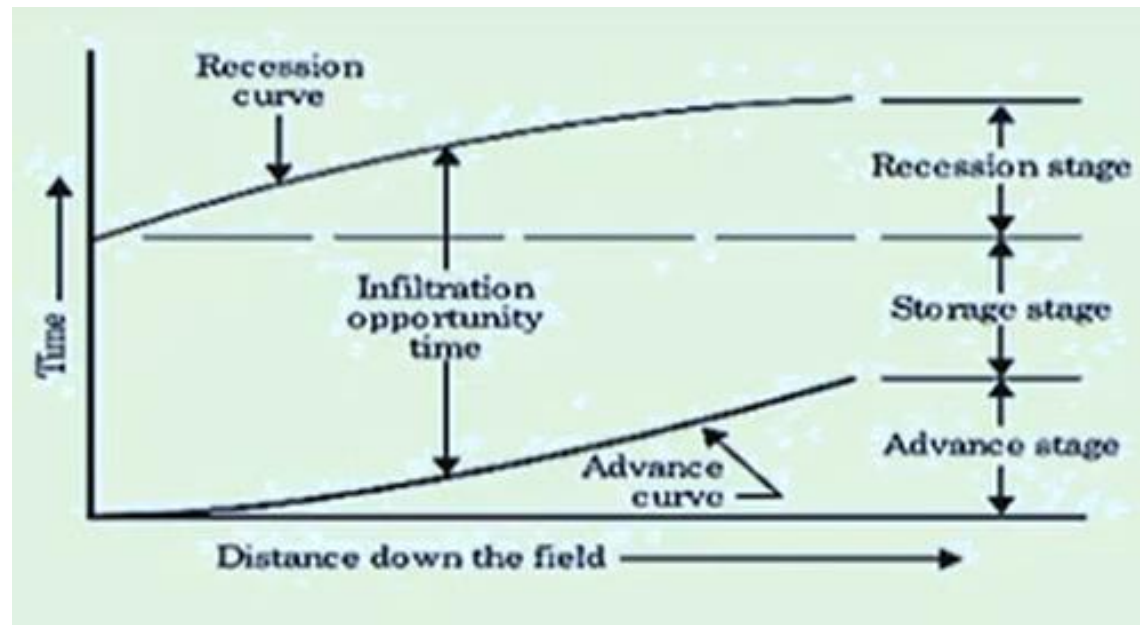
- Begins when depletion phase ends and continues until drying front reaches the end.



Phase of surface irrigation . . .

Infiltration or Intake opportunity time

- Time interval between the time it arrives at a point during the advance phase and departs during recession
- Is the vertical distance between the advance and recession curves
- For irrigation uniformity to be high: Opportunity time must be same throughout the field



Design of Surface Irrigation Systems

- The surface irrigation design process is a procedure matching the most desirable frequency and depth of irrigation and the capacity and availability of the water supply while controlling deep percolation, runoff, evaporation and operational losses.
- It first involves assessing the general topographic conditions, soils, crops, farming practices anticipated and farm operators' desires and finance for the field or farm in question.
- Properly designed and operated surface irrigation systems can enhance crop yields.
- Inadequate design and management result in excessive water losses through deep percolation and/or tail water runoff.
- Also, high efficiencies are not generally attained unless design, operation and management are of a higher standard and distribution.

Design of Surface Irrigation Systems . . .

- Is the process of making decisions concerning the values of **flow rate, Q_o , field dimensions like (length of channel, L), and time of cutoff, t_{co} .**
- Surface irrigation Design inputs are (or the performance of a surface irrigation system is expressed as):

$$P = f (I, S, n, Z_r, G, q_o, L, t_{co})$$

Where P = performance of surface irrigation

I = symbolizes the infiltration parameters

S = channel bed slope

n = hydraulic resistance

G = symbolizes geometry parameters

Z_r = required amount of application

q_o = unit flow rate at the head end of the channel

t_{co} = time of cut off

L = channel length

System or field Parameters:

are measured or assumed properties, **not manipulated or influenced by designer**

System (field decision)

Variables: can manipulated by designer

Design of Furrow Irrigation Systems

Non erosive stream size, $Q_{\max}$

- it is desirable to operate the furrow at a velocity that is non erosive, b/c to maintain proper furrow shape and reduce sediment loss from the head of the field and deposition at the tail of the field or adjacent water way.
- the maximum non-erosive stream size is:

$$Q_{\max} = \frac{0.6}{S} \text{ l/s}$$

Average depth of water application:

- The average depth of water applied during irrigation is calculated from the following relation:

$$D = \frac{Q \times 3600}{S \times L}, \text{ cm}$$

D = depth of water (cm) applied in one hr.

Q = stream flow (lit/sec)

S = furrow spacing(m)

L = furrow length(m) irrigated in one hr.

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Volume of channel storage, V_s

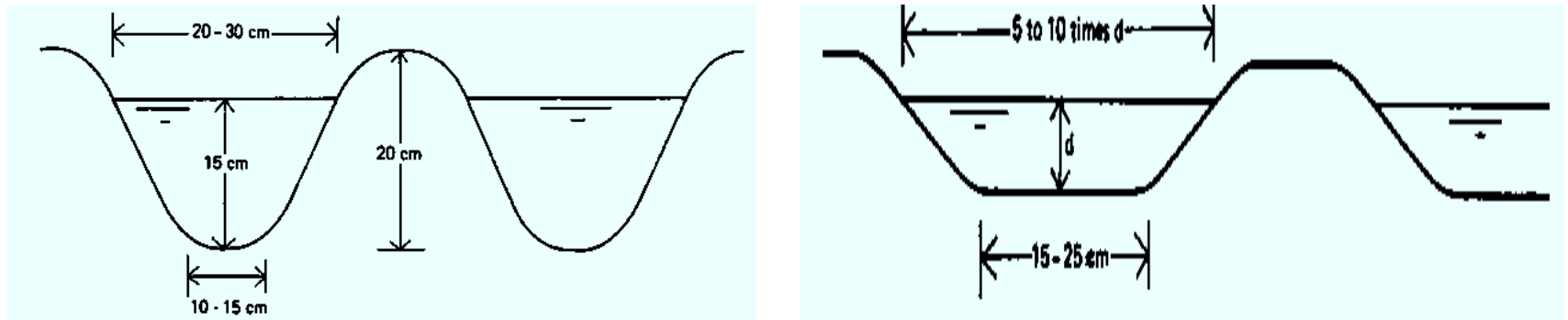
$$V_s = \frac{L}{0.305} \left[2.947 \left(\frac{Qn}{S^{0.5}} \right)^{0.735} - 0.0217 \right]$$

The adjusted wetted perimeter, P

$$P = 0.265 \left(\frac{Qn}{S^{0.5}} \right)^{0.425} + 0.227 \quad \text{where } Q = \text{volumetric inflow rate, l/s}$$

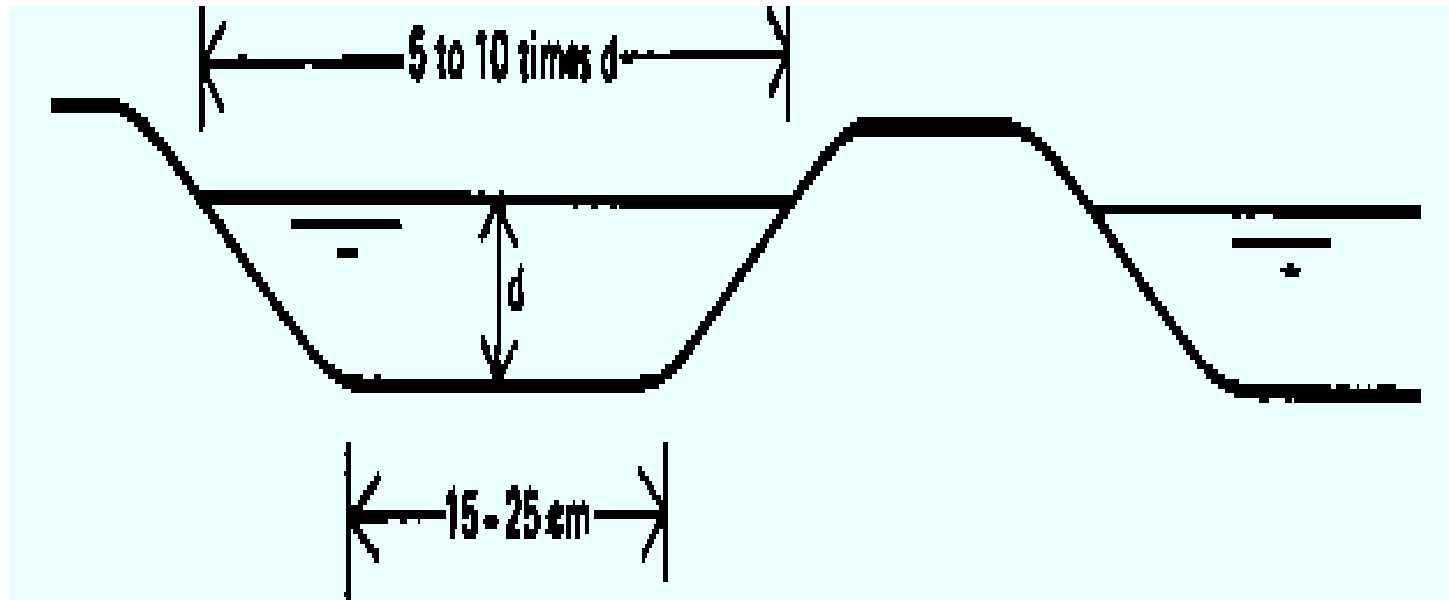
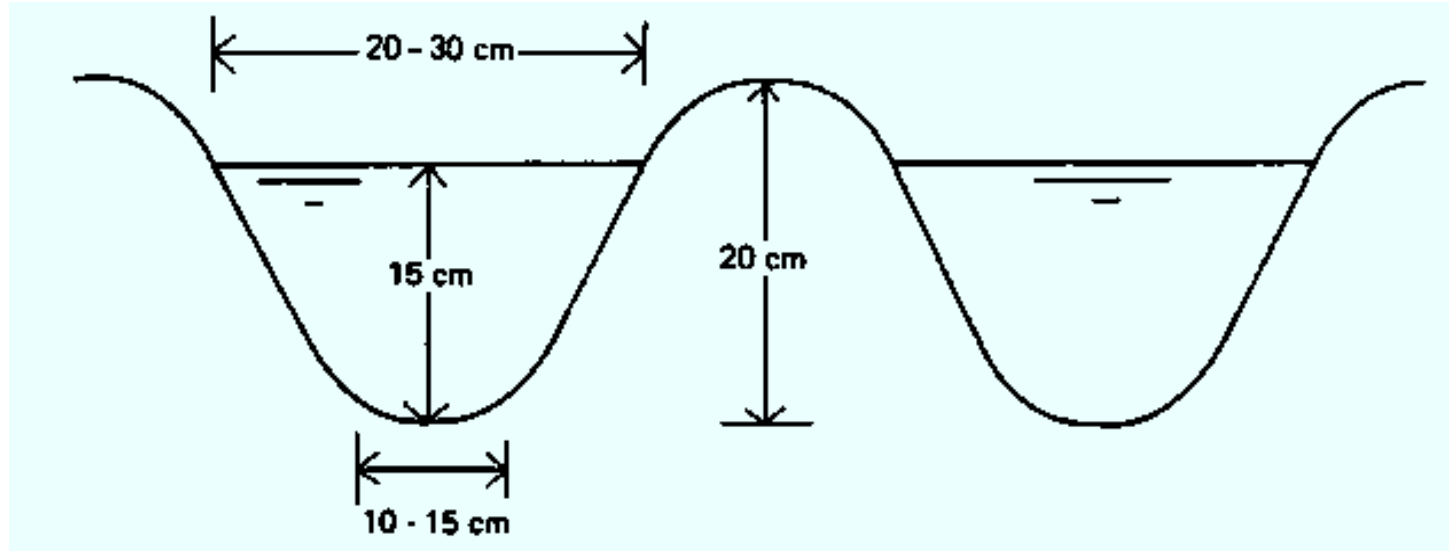
Channel geometry: Furrows, have **parabolic**, **triangular** or **trapezoidal** cross-sections. Basins and borders, generally have a wide rectangular channels.

- The usual size of furrow crop is about 25cm wide & to 10 - 15cm deep



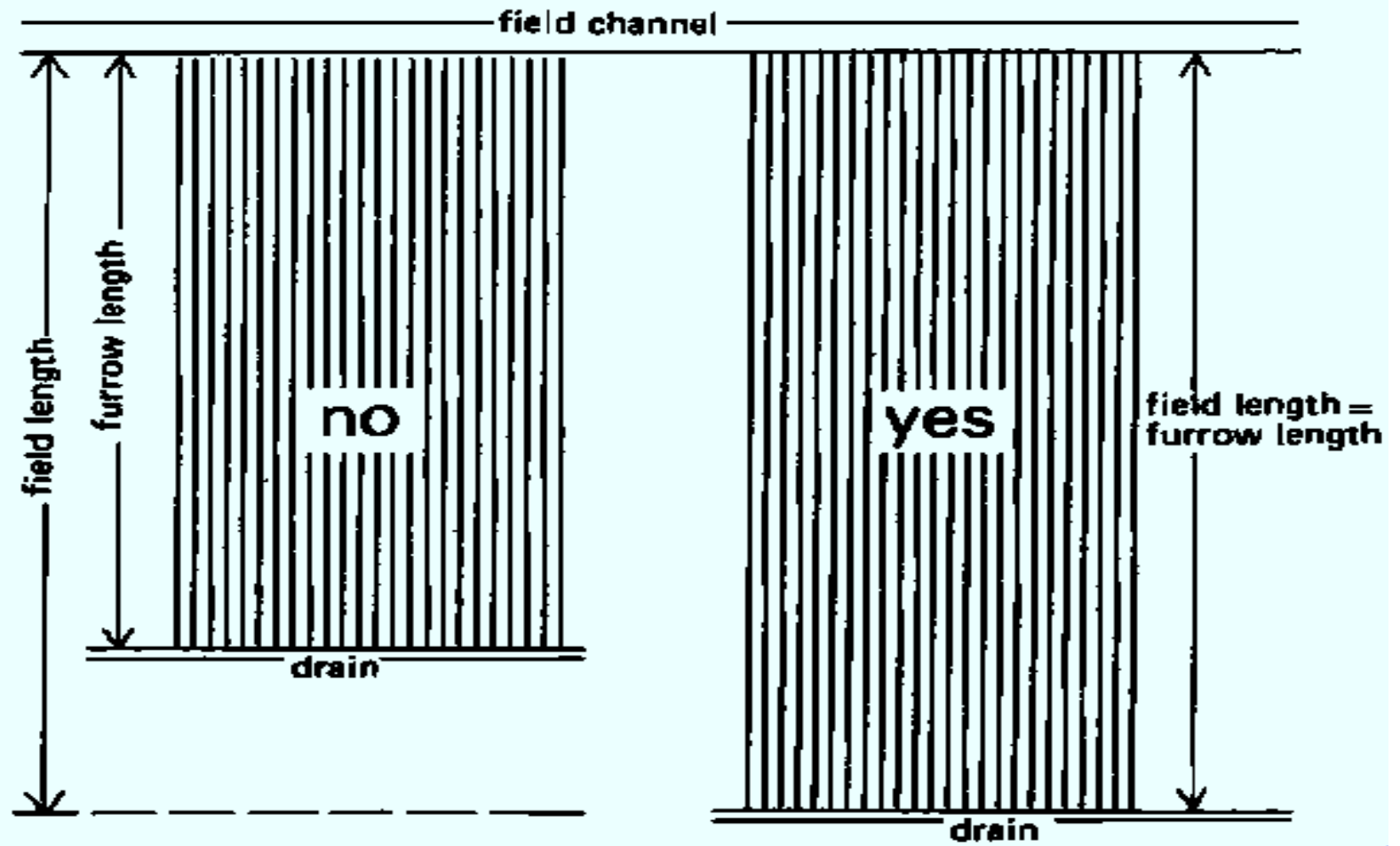
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Deep, narrow furrow on sandy soil



Wide, shallow furrow on clay soil

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Average intake over the length of the furrow (equivalent depth infiltrated over wetted surface area of field), I

$$I = \frac{1}{LP} (V_{in} - V_{out} - V_s) , \text{ mm}$$

- The **required depth of infiltration** for a furrow system must be expressed as an equivalent depth over the total field area. **Average Infiltration depth** is given by:

$$i = [at^b + C] \frac{P}{w}$$

Where a, b, C are intake family coefficients

t = time of infiltration, min

w = furrow spacing, m

P = adjusted wetted perimeter, m

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Intake family	a	b	C	f	g
0.05	0.5334	0.618	7.0	7.16	1.088×10^{-4}
0.10	0.6198	0.661	7.0	7.25	1.251×10^{-4}
0.15	0.7110	0.683	7.0	7.34	1.414×10^{-4}
0.20	0.7772	0.699	7.0	7.43	1.578×10^{-4}
0.25	0.8534	0.711	7.0	7.52	1.741×10^{-4}
0.30	0.9246	0.720	7.0	7.61	1.904×10^{-4}
0.35	0.9957	0.729	7.0	7.70	2.067×10^{-4}
0.40	1.064	0.736	7.0	7.79	2.230×10^{-4}
0.45	1.130	0.742	7.0	7.88	2.393×10^{-4}
0.50	1.196	0.748	7.0	7.97	2.556×10^{-4}
0.60	1.321	0.757	7.0	8.15	2.883×10^{-4}
0.70	1.443	0.766	7.0	8.33	3.209×10^{-4}
0.80	1.560	0.773	7.0	8.50	3.535×10^{-4}
0.90	1.674	0.779	7.0	8.68	3.862×10^{-4}
1.00	1.786	0.785	7.0	8.86	4.188×10^{-4}
1.50	2.284	0.799	7.0	9.76	5.819×10^{-4}
2.00	2.753	0.808	7.0	10.65	7.451×10^{-4}

Design of Furrow Irrigation Systems . . .

Advance time for stream of water moving down the furrow, T_t

$$T_t = \frac{x}{f} \exp \left[\frac{gx}{Qs^{0.5}} \right]$$

Where T_t = advance time, min

x = distance down the furrow, m

f, g = advance coefficient (Table)

Q = volumetric inflow rate, l/s

S = furrow slope, m/m

Cut-off time, T_{co}

- reflects an irrigation management **decision made by the farmer and designer**.
- It should be an adequate length of time to infiltrate a satisfactory depth of water over the length of the furrow without causing excessive deep percolation.
- is normally set equal to the **time to advance, T_t** to the end of the furrow **plus** the **required net infiltration time, T_n** less recession time.

$$T_{co} = T_t + T_n$$

Cont'd . . .

Net infiltration time, T_n : is the required infiltration time.

- By letting infiltration depth in equal with the desired net depth of infiltration (i.e. $i = i_n$), i_n is determined by:

$$T_n = \left[\frac{i_n \left(\frac{W}{P} \right) - c}{a} \right]^{1/b}$$

Infiltration Opportunity time, T_o

- is equal to the time of water application minus the advance time plus the recession time.

$$T_o = T_{co} - T_t + T_r$$

- The recession time is assumed **Zero** for open-ended gradient furrows (i.e for furrows whose slope is not equal to zero) with out loss of accuracy.

For open-ended gradient furrows,

$$T_o = T_{co} - T_t$$

Design of Furrow Irrigation Systems . . .

The **average infiltration opportunity time over distance x** down the furrow is given by:

$$T_{o-x} = T_{co} - \frac{0.0929}{f(x) \left[\frac{0.305\beta}{x} \right]^2} [(\beta - 1)\exp(\beta) + 1]$$

$$\text{Where } \beta = \frac{gx}{QS^{0.5}}$$

- The average infiltration time for the full furrow length, T_{o-L} , is obtained by substituting L , in to the above eqⁿ.
- The average depth of infiltration for the entire furrow length, i_{avg} , is therefore determined by substituting T_{o-L} in to the equation of i for t .

$$i_{avg} = [aT_{o-L}^b + c] \frac{P}{W}$$

Cont'd . . .

Gross depth of water application, I_g

- is defined as the required net depth of irrigation, i_n , divided by the product of the application and distribution efficiencies.

$$I_g = \frac{I_n}{E_a E_d}$$

- If evaporation is neglected, $E_a \approx 100\%$ $I_g = \frac{I_n}{E_d}$
- The equivalent gross depth of application as a function of inflow rate and field geometry, is

$$I_g = \frac{60(Q)(T_{co})}{wL}$$

where I_g = gross depth of application in mm
 Q = inflow rate, l/s
 w = furrow spacing, m

The **equivalent depth of surface runoff**, d_{ro} $d_{ro} = I_g - i_{avg}$

The **equivalent depth of deep percolation**, d_{dp} $d_{dp} = I_{avg} - i_n$

Design of Furrow Irrigation Systems . . .

Furrow slope (%)	Maximum stream size (l/s) per furrow	Clay		Loam		Sand	
		Net irrigation depth (mm)					
		50	75	50	75	50	75
0.0	3.0	100	150	60	90	30	45
0.1	3.0	120	170	90	125	45	60
0.2	2.5	130	180	110	150	60	95
0.3	2.0	150	200	130	170	75	110
0.5	1.2	150	200	130	170	75	110

Design of Irrigation Systems . . .

Reading assignment

1. Adjusted wetted perimeter under the cut – back
2. Level Basin System Design
3. Graded Border System Design

Example

Example: Given the following information,

Intake family, $IF = 0.3$

Furrow length, $L = 275\text{m}$

Furrow slope, $s = 0.004 \text{ m/m}$

Roughness coefficient, $n = 0.04$

Net irrigation depth, $i_n = 75\text{mm}$

Inflow rate, $Q = 0.6 \text{ l/s}$

Furrow spacing, $W = 0.75\text{m}$

Neglect evaporation loss, $E_a = 1$

Compute

- the desired time to cut – off, T_{co}
- the equivalent depths of surface run off and deep percolation, d_{ro} , d_{dp} ,
- the distribution pattern efficiency.

Example

Solution:

Intake family, $IF = 0.3 \Rightarrow a = 0.9246, b = 0.72, c = 7.0, f = 7.61$
 $g = 1.904 \times 10^{-4}$

Advance time, T_t

$$T_t = \frac{x}{f} \exp \left[\frac{gx}{QS^{0.5}} \right] = \frac{275\text{m}}{7.61} \exp \left[\frac{(1.904 \times 10^{-4})(275)}{(0.6)(0.004)^{0.5}} \right] = 144 \text{ min}$$

Adjusted wetted perimeter, P

$$P = 0.265 \left(\frac{Qn}{S^{0.5}} \right)^{0.425} + 0.227 = 0.265 \left(\frac{0.6 \times 0.04}{(0.004)^{0.5}} \right)^{0.425} + 0.227 = 0.40\text{m}$$

Net Infiltration time, T_n

$$T_n = \left[\frac{i_n \left(\frac{w}{P} \right) - c}{a} \right]^{1/b} = \left[\frac{75\text{mm} \left(\frac{0.75\text{m}}{0.40\text{m}} \right) - 7.0}{0.9246} \right]^{1/0.72} = 999\text{min}$$

Example

Solution:

- a) Design cut-off time, T_{co}

$$T_{co} = T_t + T_n = 144 + 999 = 1143 \text{ min (19.05 hr)}$$

- b) Gross application depth, I_g

$$I_g = \frac{60(Q)(T_{co})}{wL} = \frac{60(0.6l/s)(1143min)}{0.75m \times 275m} = 199.51 \approx 200\text{mm}$$

Average infiltration opportunity time over distance L, T_{o-L}

$$T_{o-x} = T_{co} - \frac{0.0929}{f(x) \left[\frac{0.305\beta}{x} \right]^2} [(\beta - 1)\exp(\beta) + 1]$$

$$\text{Where } \beta = \frac{gx}{QS^{0.5}} = \frac{(1.904 \times 10^{-4})(275)}{(0.6)(0.004)^{0.5}} = 1.37971 \approx 1.38$$

$$T_{o-L} = 1143 - \frac{0.0929}{7.61 \times 275 \left[\frac{0.305 \times 1.38}{275} \right]^2} [(1.38 - 1)\exp(1.38) + 1]$$

$$T_{o-L} = 1143 - 47.6 \approx 1095 \text{ min}$$

Solution:

Example

Average infiltration depth, i_{avg}

$$i_{avg} = [aT_{o-L}^b + C] \frac{P}{W} = [0.9246 \times 1095^{0.72} + 7.0] \frac{0.4}{0.75}$$

$$i_{avg} \approx 80 \text{ mm}$$

$\therefore$ Hence the average infiltration depth along the furrow is 80mm

Surface runoff depth, d_{ro}

$$d_{ro} = I_g - i_{avg} = 200 \text{ mm} - 80 \text{ mm} = 120 \text{ mm}$$

$\therefore$ Hence the amount of water lost from the furrow is 120mm

Deep percolation depth, d_{dp}

$$d_{dp} = i_{avg} - i_n = 80 \text{ mm} - 75 \text{ mm} = 5 \text{ mm}$$

$\therefore$ Hence the amount of water drained below crop root zone is 5 mm

c) **Distribution pattern efficiency, E_d**

$$E_d = \frac{i_n}{I_g} \times 100\% = \frac{75 \text{ mm}}{200} \times 100\% = 37.5\%$$

Pressurized Irrigation Methods

- Water moves through the pipes under pressure
- Energy is required in order to develop enough head to overcome frictional resistances in pipes & pump section.
- This method saves much more water than traditional surface flow irrigation, where water from canals or bodies of water is released into open furrows or basins and distributed by gravity.

Major components of pressurized system

-  Water sources
-  Energy sources
-  Distribution Network-pipe system & emission devices

Pressurized Irrigation Methods . . .

Advantages

- ⇒ Avoid the water loss of open surface flow irrigation.
- ⇒ Higher water application efficiency
- ⇒ Especially valuable in arid and semi-arid areas.
- ⇒ When the land topography is irregular, the land gradient is steeper, and soil is easily erodible, hence unsuitable for surface irrigation.
- ⇒ When the land soil is excessively permeable, so as not to permit good water distribution by surface irrigation; or when the soil is highly impermeable.
- ⇒ When the water table is high.

Pressurized Irrigation Methods . . .

Disadvantages

- ⇒ Water quality is crucial to the prolonged operation of pressurized irrigation systems
- ⇒ Dissolved minerals leads to frequent blocking of emitters
- ⇒ Routine maintenance is needed to unblock delivery fittings and to maintain pumps and fertigation units
- ⇒ Water needs to be filtered before entering the irrigation system to remove suspended solids such as silt, algae, sand and dust, which will clog emitters
- ⇒ High initial investment as compared to surface irrigation
- ⇒ Wind distorts sprinkler pattern and causes uneven distribution of water
- ⇒ Ripened soft fruits may be affected by spraying water

Pressurized Irrigation Methods . . .

1. Sprinkler Irrigation

- Is a method of applying water in the form of a **spray** at a rate less than infiltration rate of the soil through a **network of pipes and pump**
- It is a kind of an **artificial rain** and therefore, gives **very good results**.
- Water should be clean
- It is costly process of irrigation
- Suitable to **most crops, any slope, sandy soils**
- Based on arrangement for spraying, sprinklers can be:
 - › Fixed nozzle,
 - › Perforated or rotating type

Pressurized Irrigation Methods . . .



Sprinkler Irrigation: artificial rain

Pressurized Irrigation Methods . . .



Sprinkler Irrigation: artificial rain

Pressurized Irrigation Methods . . .



Sprinkler Irrigation: artificial rain

1. Sprinkler Irrigation . . .

Components of sprinkler system:

- © **Pressure generating units** (Pump unit): The pump unit is usually a centrifugal pump which takes water from the source and provides adequate pressure for delivery into the pipe system
- © **Water carrier units** (Mainlines , sub mainlines, Laterals)
- © **Water delivery units** (riser pipes and Sprinkler nozzle)
- © **Quality improvement sub units** (Screens , Desilting basins)
- © **Ancillary units** (Fertilizer and other chemical applicator)

1. Sprinkler Irrigation . . .

Components of sprinkler system:

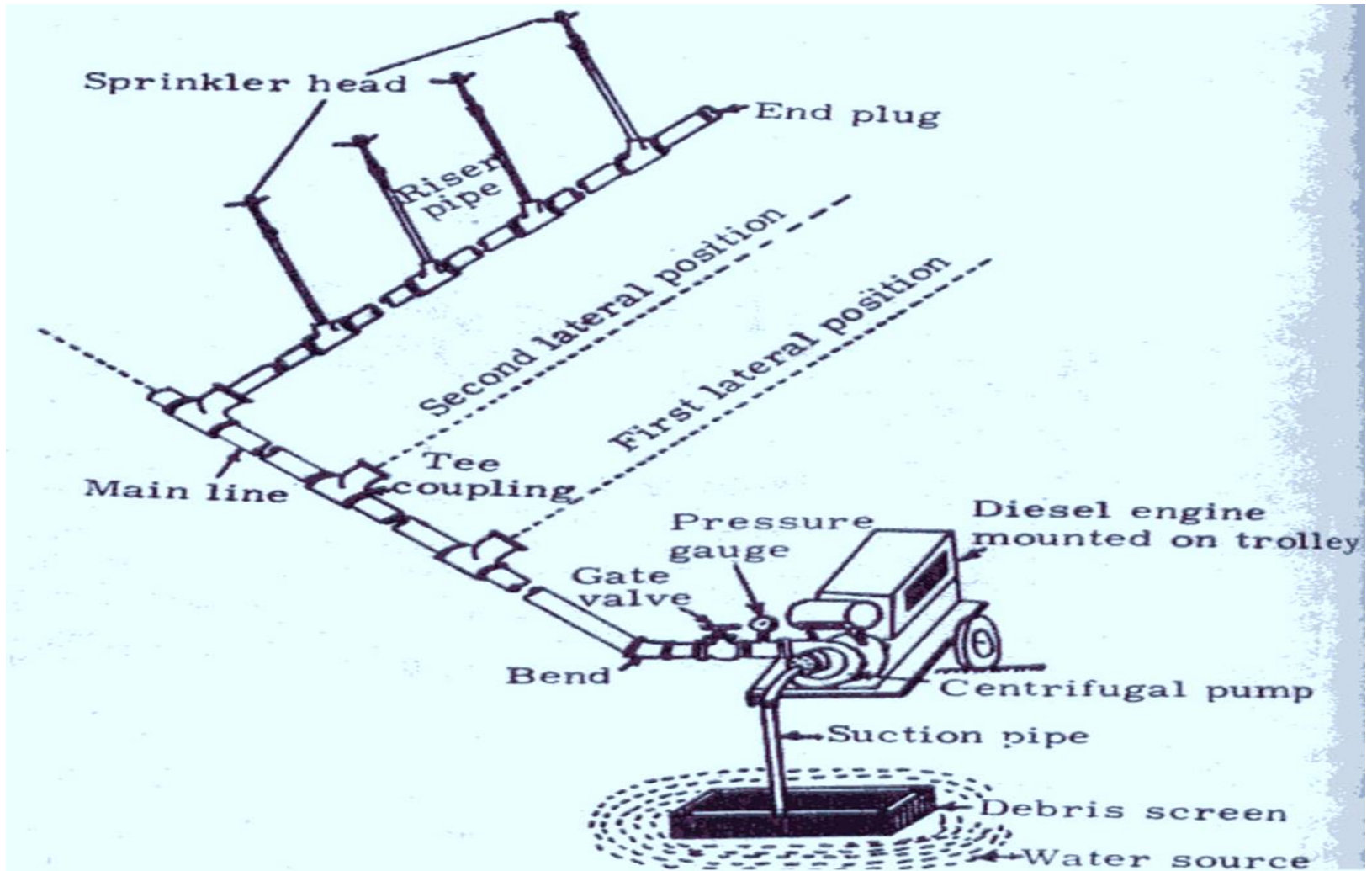
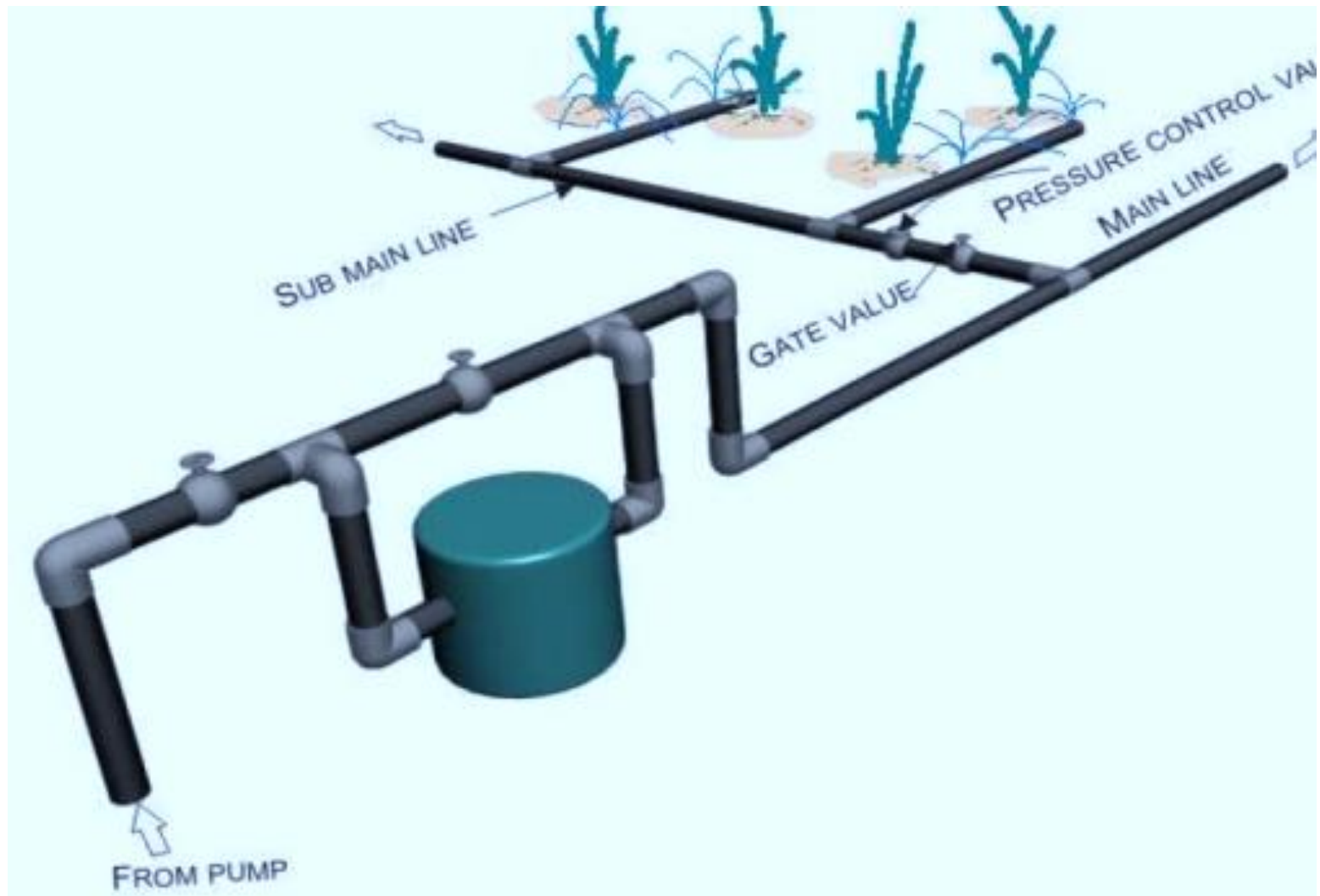


Fig. Components of sprinkler irrigation system

1. Sprinkler Irrigation . . .

Components of sprinkler system:



Components of sprinkler system

1. Sprinkler Irrigation . . .

- This method is more useful where:
 - The land cannot be prepared for surface methods,
 - Slopes are excessive,
 - Topography is irregular,
 - Soil is erosive,
 - Soil is excessively permeable.
- Sprinkler system can be further sub divided into **three** heads:
 - i. Permanent system
 - ii. Semi-permanent system
 - iii. Portable system

1. Sprinkler Irrigation . . .

Wetting Patterns:

- The main objective of a sprinkler system is to apply water as uniformly as possible to fill the root zone of the crop with water.
- The wetting pattern from a single rotary sprinkler is not very uniform.
- Normally the area wetting is circular. The heaviest wetting is close to the sprinkler.

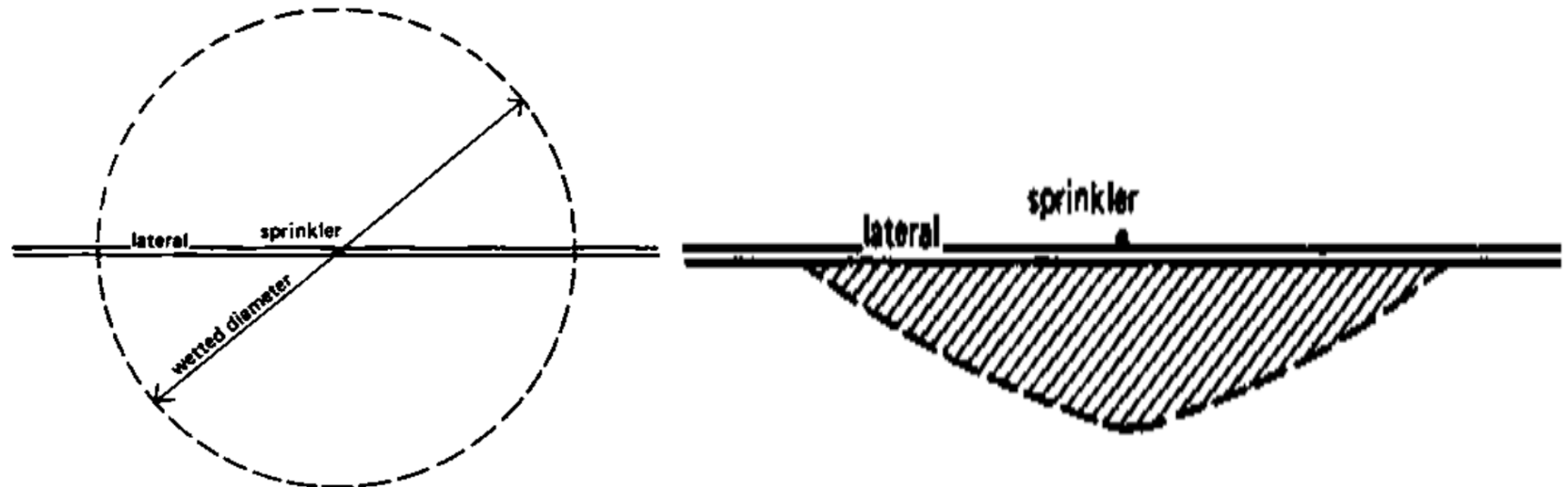
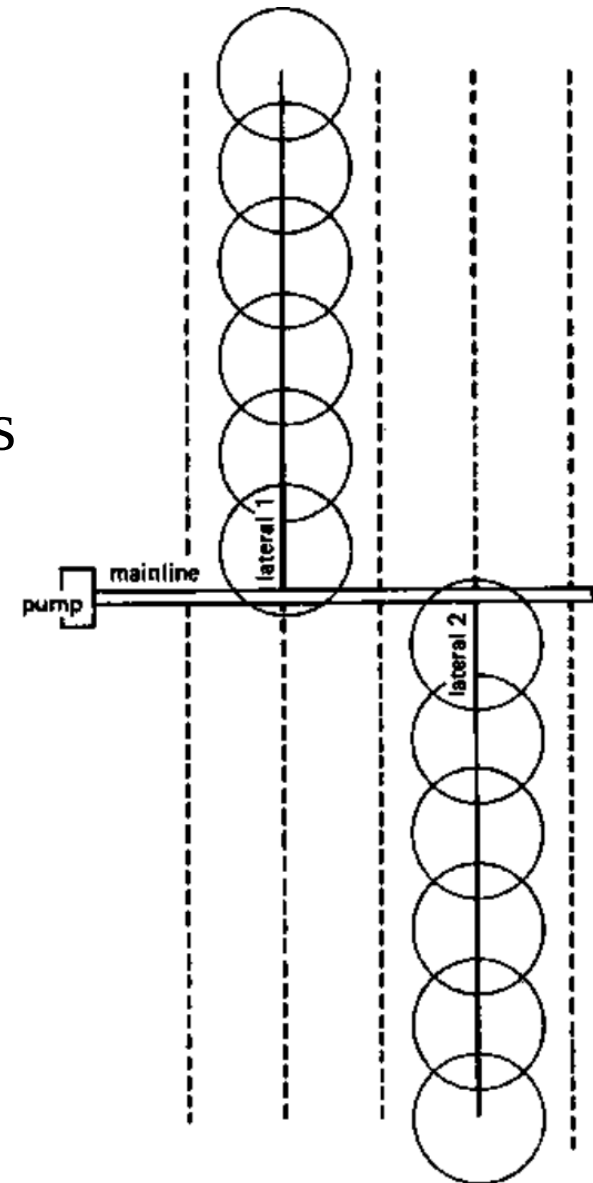


Figure Wetting pattern for a single sprinkler (Top & Side View)

1. Sprinkler Irrigation . . .

Wetting Patterns . . .

- For good uniformity several sprinklers must be operated close together so that their patterns overlap.
- The uniformity of sprinkler applications can be affected by wind and water pressure.
- To reduce the effects of wind the sprinklers can be positioned more closely together.
- For good uniformity the overlap should be at least 65% of the wetted diameter.



1. Sprinkler Irrigation . . .

Application rate

- This is the average rate at which water is sprayed onto the crops and is measured in mm/hour.
- The application rate depends on
 - the size of sprinkler nozzles
 - the operating pressure
 - the distance between sprinklers

Sprinkler drop sizes

- Water sprays from a sprinkler it breaks up into small drops between 0.5 and 4.0 mm in size.
- The small drops fall close to the sprinkler whereas the larger ones fall close to the edge of the wetted circle.
- Drop size is also controlled by pressure and nozzle size.
- When the pressure is low, drops tend to be much larger as the water jet does not break up easily.

1. Sprinkler Irrigation . . .

Limitation of sprinkler Irrigation:

- Wind distorts sprinkler pattern and causes uneven distribution of water
- Ripened soft fruits may be affected by spraying water.
- Water must be clean and free of sand, & debris. If not, nozzle blockage may happen.
- High initial investment as compared to surface irrigation.
- High power requirements
- Fine textured soils with slow infiltration rate can not be irrigated efficiently.

Pressurized Irrigation Methods . . .

2. Drip Irrigation

- It is sometimes called *trickle irrigation*
- Involves **dripping water onto the soil** (close to the plant roots) **at very low rates** from a system of small diameter plastic pipes fitted with outlets called *emitters or drippers*
- Most efficient but most expensive method and hence applicable in water scarce areas.
- Water is applied directly to the crop i.e. entire field is not wetted. Water is conserved
- Weeds are controlled because only the places getting water can grow weeds.
- There is a low-pressure system.

2. Drip Irrigation . . .

- There is reduced evaporation, only potential transpiration is considered.
- There is no need for a drainage system.
- Water spreads laterally and vertically.
- Water with sediment impurities should not be used for drip as it may cause clogging of emitters
- It is more economical for orchards & unsuitable for closely planted crops like wheat & barely



2. Drip Irrigation . . .



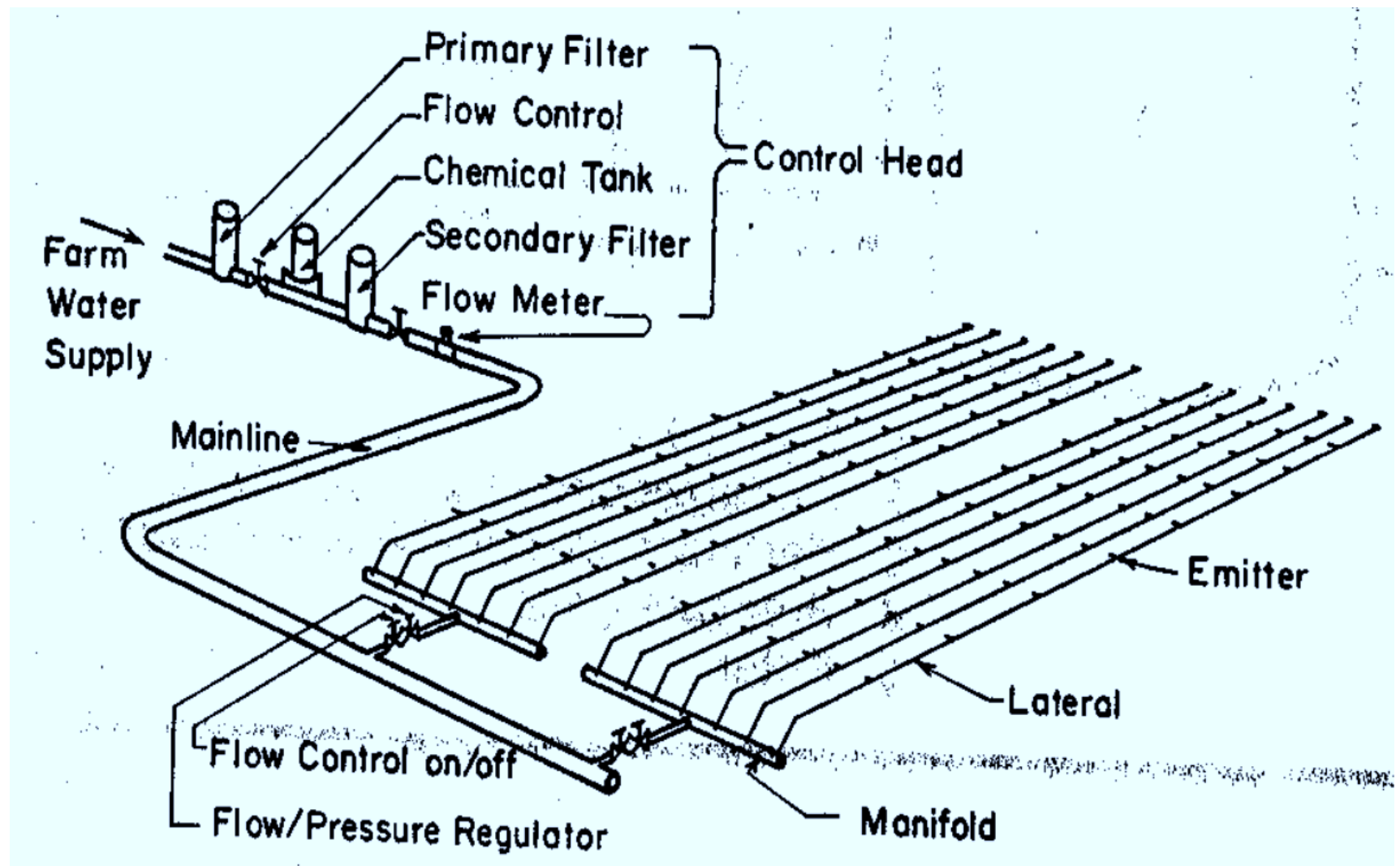
2. Drip Irrigation . . .

Components of Drip system:

- © **Pump unit:** takes water from the source and provides the right pressure for delivery into the pipe system.
- © **Control head:** consists of valves to control the discharge and pressure in the entire system. It may also have filters to clear the water.
- © **Mainlines, sub mains and laterals** supply water from the control head into the fields. They are usually made from PVC or polyethylene hose .
- © **Emitters or drippers** are devices used to control the discharge of water from the lateral to the plants.
 - They are usually spaced more than 1 meter apart with one or more emitters used for a single plant such as a tree.
 - For row crops more closely spaced emitters may be used to wet a strip of soil.

2. Drip Irrigation . . .

Components of Drip system:



2. Drip Irrigation . . .

Suitable crops

- Drip irrigation is most suitable for row crops (vegetables, soft fruit), tree and vine crops where one or more emitters can be provided for each plant.
- Generally only high value crops are considered because of the high capital costs of installing a drip system.

Suitable slopes

- Drip irrigation is adaptable to any farmable slope.

Suitable soil

- Drip irrigation is suitable for most soils.
- On clay soils water must be applied slowly to avoid surface water ponding and runoff.
- On sandy soils higher emitter discharge rates will be needed to ensure adequate lateral wetting of the soil.

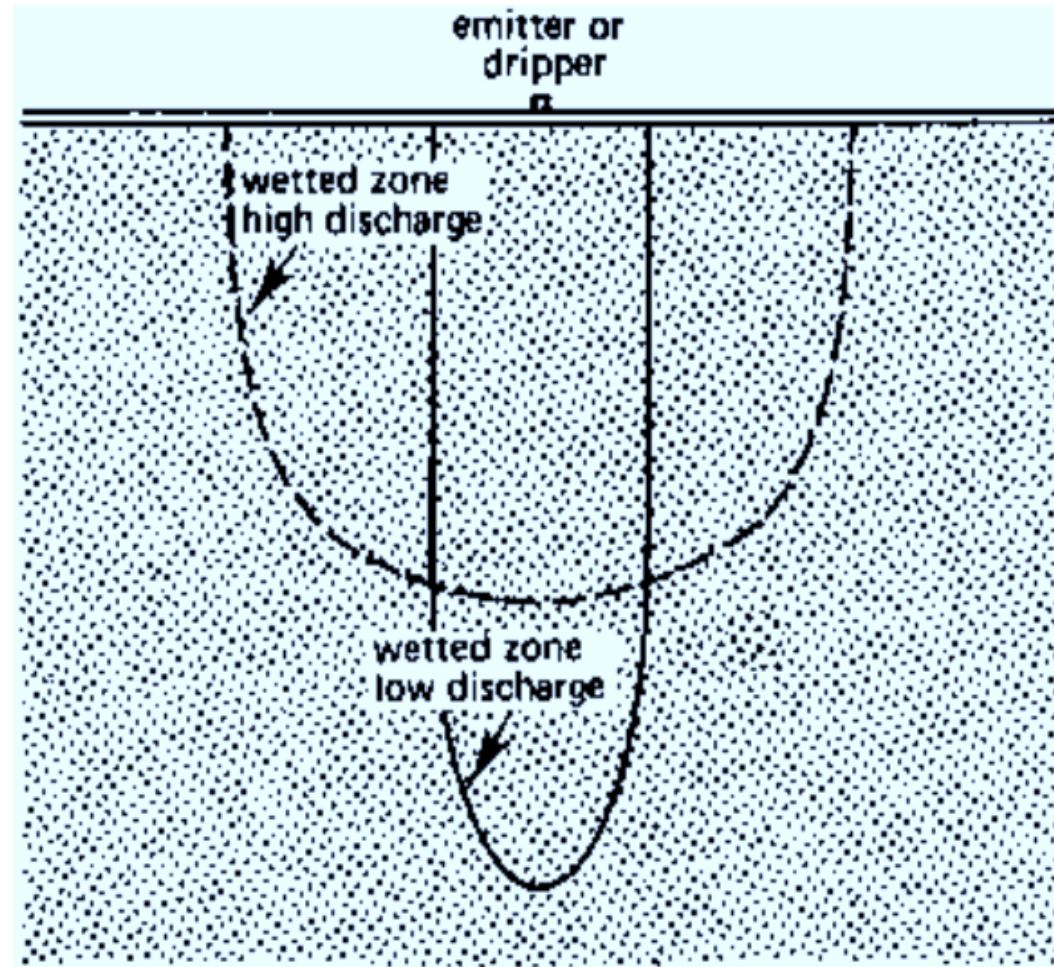
Suitable irrigation water

- One of the main problems with drip irrigation is blockage of the emitters.
- All emitters have very small waterways ranging from 0.2-2.0 mm in diameter and these can become blocked if the water is not clean.

2. Drip Irrigation . . .

Wetting Patterns . . .

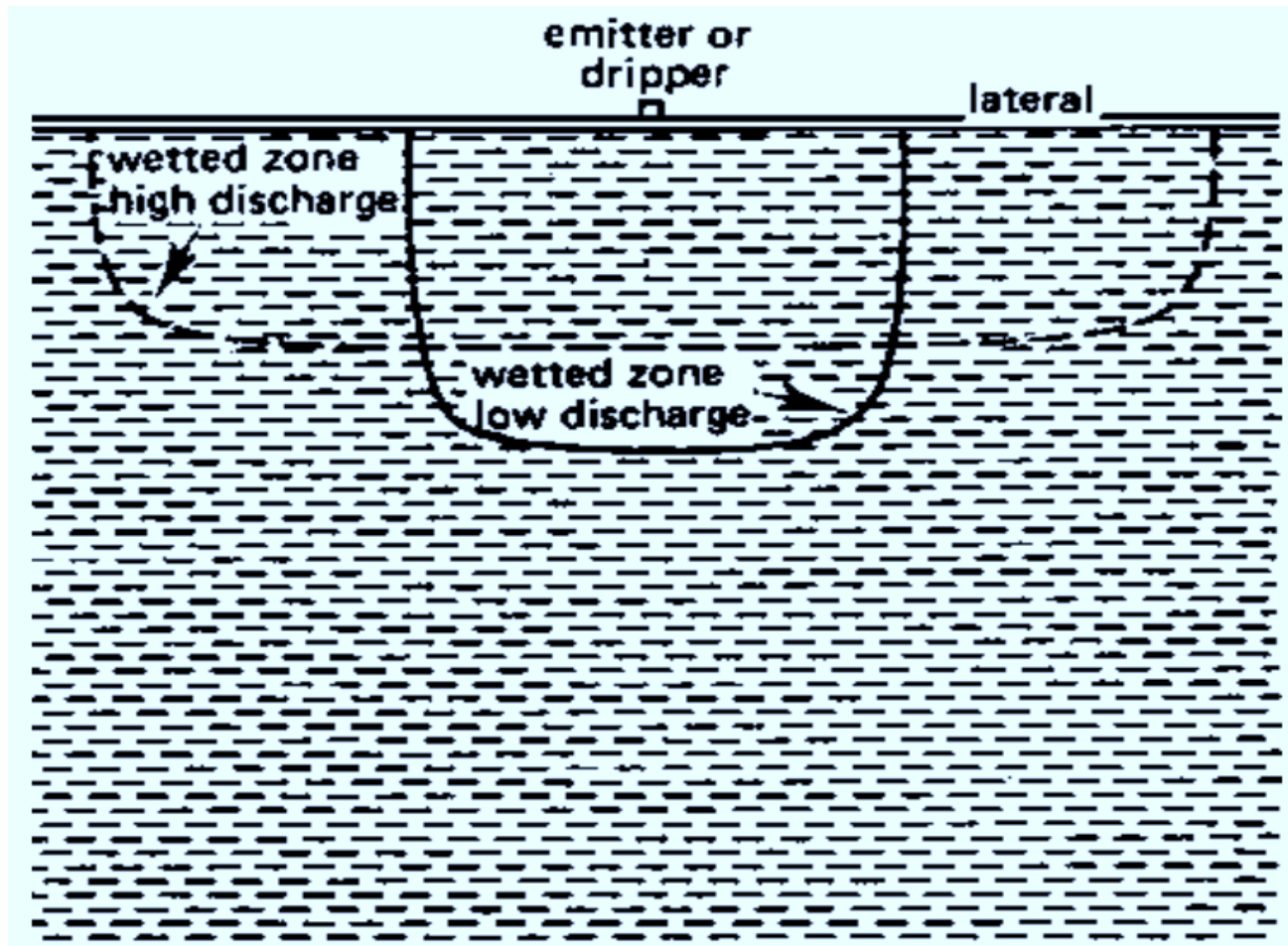
- Unlike surface & sprinkler irrigation, drip irrigation **only wets part of the soil root zone**. This may be as low as 30% of the volume of soil wetted by the other methods.
- The wetting patterns which develop from dripping water onto the soil **depend on discharge and soil type**.



Wetting patterns for sand soils with high and low discharge rates

2. Drip Irrigation . . .

Wetting Patterns . . .



Wetting patterns for clay soils with high
and low discharge rates

2. Drip Irrigation . . .

Advantages of drip irrigation

- ⇒ High application efficiency (90%). Hence, saves water
- ⇒ Requires lower line pressure compared to sprinkler. Saves energy requirement
- ⇒ Fertilizer application with high precision
- ⇒ Elimination of the need for Drainage
- ⇒ Possible to irrigate marginal soils

Limitations of drip irrigation

- High initial investment
- Requires clearer water
- Salt accumulation at the periphery- no continuous leaching
- Pipes are liable to mechanical damages
- Limited root development- localized to the wetted area.

Land Grading/Levelling

- Reshaping of the field surface to a planned grade in order to control the flow of water is known as **land levelling**
- It involves **site clearing, removal of abrupt irregularities, filling of depressions & moving large quantity of earth over considerable distance and**
- It involves moving soil in order to have **level fields for basin irrigation** or **uniform sloping fields for furrow or border strip irrigation.**
- When levelling or grading land, one should **avoid large volumes of cut and fill.**
- Besides being expensive, too much soil movement tends to leave shallow topsoil in areas of cut, which is not ideal for crop production.

Land Grading/Levelling

- When uneven land is irrigated, the high spots are watered too little and the low spots too much.
- Due to this, uneven crop growth, yield reduction, and loss of water is occurred.

❖ Importance of land levelling

- ⇒ To enable efficient surface irrigation
- ⇒ To provide surface drainage or removal of surface water
- ⇒ To control the flow of water,
- ⇒ To check soil erosion
- ⇒ To have uniform distribution of water
 - › Low areas may receive excess water – water logging
 - › High areas may not receive sufficient water – low yields

Land Grading/Levelling . . .



Land Grading/Levelling . . .

- Land leveling operations may be grouped into three phases:
 - ∴ Rough grading
 - ∴ Land leveling
 - ∴ Land smoothing

Rough grading

Is the removal of **abrupt irregularities** such as mounds, dunes and rings, and filling of pits, depressions and gullies

Land leveling

Land grading = land forming = land shaping

It requires moving large quantities of earth over considerable distance

Land smoothing

Small irregularities are removed and a plane surface obtained by land smoothing which is the final operation in land leveling.

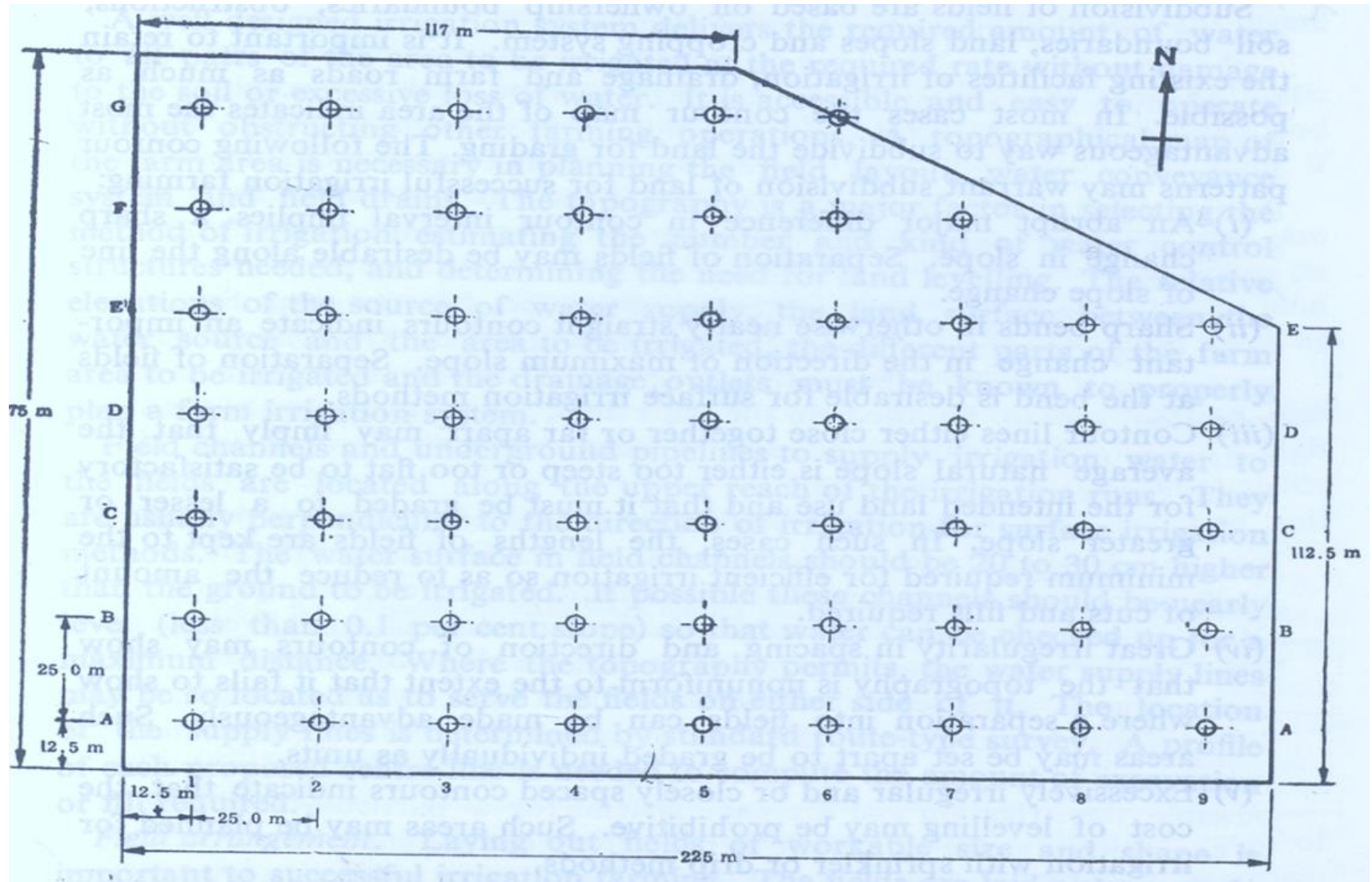
Land Grading/Levelling . . .

❖ Land Grading Survey and Design

- A detailed topographic survey, preferably grid, is needed to calculate the most economic land levelling requirements.
- Prior to leveling design, the following points should be known
 - › the location of the field boundaries,
 - › irrigation water supply system,
 - › drainage systems and the farm roads
- After field boundaries are established, a **grid systems are established** and **stakes at the grid points are set**.
- The usual grid space is 25m×25m, but 30m×30m, 20m×20m, and 15m×15m are also used
- Based on the spot heights of the grid points and the required gradient of the land, the cut and fill can be calculated.
- For convenience, the **row lines** are **lettered** and the **column lines** are **numbered**

Land Grading/Levelling . . .

Land Grading Survey and Design . . .



Land Grading/Levelling . . .

❖ Land leveling design methods.

- The basic methods of land leveling design (for calculating the amounts of soil cuts and fills) are:

⚡ Plane method

⚡ Profile method

⚡ Plan-inspection method and

⚡ Contour adjustment method

Land Grading/Levelling . . .

Plane Method

- Is the most commonly used method of land leveling design.
- The **procedures for plane method**:
 - i. Complete the design and construction survey
 - ii. Determine the initial elevation at each grid point (E_i)
 - iii. Subdivide the area into sub-areas, each of which can be leveled to a plane surface
 - iv. Determine the weighted average elevation of the centroid of the field

$$E_{avg} = \frac{\sum E_i}{n} \quad \text{where } E_i = \text{elevation of each grid point}$$

n = number of grid points

Land Grading/Levelling . . .

✧ Plane Method . . . Cont'd

v. Determining the centroid of the field

To give equal cut and fill, the plane must pass through the centroid

$$X_c = \frac{\sum(D_x S_y)}{n} \quad Y_c = \frac{\sum(D_y S_x)}{n}$$

X_c = centroid of the field along x-direction (East-West)

D_x = distance along x-direction (East-West)

S_y = Number of stakes along y-direction (North-South)

Y_c = centroid of the field along y-direction (North-South)

D_y = distance along y-direction (North-South)

S_x = Number of stakes along x-direction (East-West)

Land Grading/Levelling . . .

✦ Plane Method . . . Cont'd

- iii. **Compute the slope of the plane of best fit**
 - Best fit slope may be as **play ground** (zero slope) or **sloped downfield**
- iv. **Compute of formation levels (new elevations), cuts or fills depths or volume**
- v. **Determination of cut & fill ratio**

Cut/fill ratios for various soils

Soils	Cut/Fill ratio
Coarse-textured soils (sandy)	1.1:1 to 1.2:1 or 110 to 120%
Medium-textured soils (clay-loam)	1.2:1 to 1.3:1 or 120 to 130%
Fine-textured soils (clayey)	1.3:1 to 1.4:1 or 130 to 140%
Organic soils	1.7:1 to 2.0:1 or 170 to 200%

Land Grading/Levelling . . .

❖ Earth-work volume

Calculate the earth-work volume

- Knowing the initial and new elevation, we can determine the cut and fill in each grid square and can calculate the total volume of soil to be moved.

$$V = \sum C \times A$$

where;

V = volume of soil to be moved (m^3).

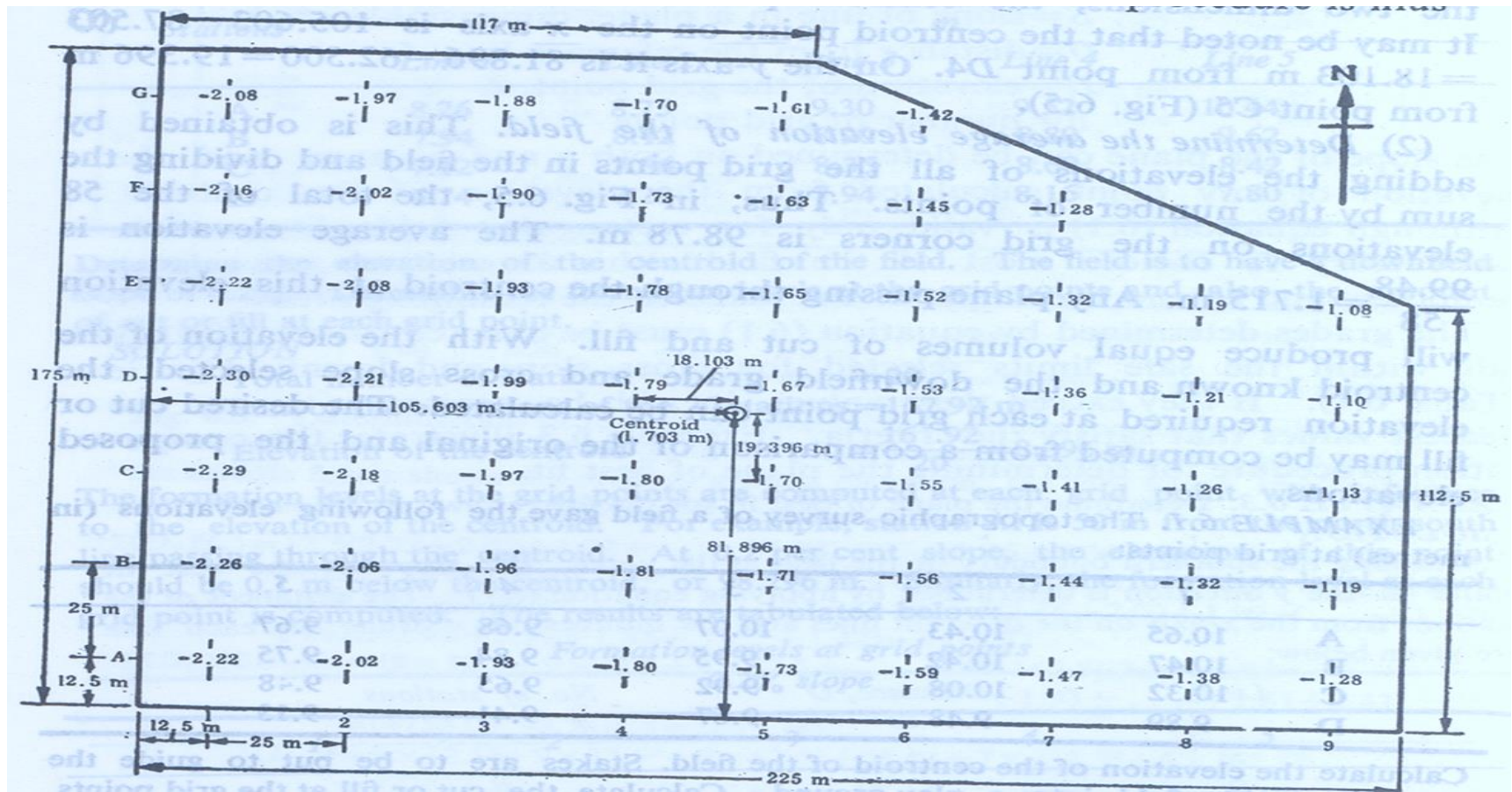
$\sum C$ = sum of all cuts (m) ($C = E_i - E_n > 0$).

A = area of grid square (m^2).

Cont'd...

Example 1:

The elevations of grid points selected at 25 m interval are determined from a topographic survey for land leveling program. Determine the average field elevation and the centroid of the field.



Cont'd...

Solution

	1	2	3	4	5	6	7	8	9
G	2.08	1.97	1.88	1.7	1.61	1.42			
F	2.16	2.02	1.9	1.73	1.63	1.45	1.28		
E	2.22	2.08	1.93	1.78	1.65	1.52	1.32	1.19	1.08
D	2.3	2.21	1.99	1.79	1.67	1.53	1.36	1.21	1.1
C	2.29	2.18	1.97	1.8	1.7	1.55	1.41	1.26	1.13
B	2.26	2.06	1.96	1.81	1.71	1.56	1.44	1.32	1.19
A	2.22	2.02	1.93	1.8	1.73	1.59	1.47	1.38	1.28

- Step 1: Determining the average elevation of the field

$$E_{avg} = \frac{\sum E_i}{n}$$

$$\sum E_i = 98.78m \quad \text{and } n = 58$$

$$E_{avg} = \frac{98.78m}{58}$$

$$E_{avg} = 1.703m$$

Cont'd...

Step 2: Determining the centroid of the field

Line	Distance, D_x	No stakes, S_y	Product
1	12.5	7	87.5
2	37.5	7	262.5
3	62.5	7	437.5
4	87.5	7	612.5
5	112.5	7	787.5
6	137.5	7	962.5
7	162.5	6	975
8	187.5	5	937.5
9	212.5	5	1062.5
Total		58	6125

$$X_c = \frac{\sum D_x S_y}{n} = \frac{6125m}{58}$$

$$X_c = 105.603m$$

Line	Distance, D_y	No stakes, S_x	Product
A	12.5	9	112.5
B	37.5	9	337.5
C	62.5	9	562.5
D	87.5	9	787.5
E	112.5	9	1012.5
F	137.5	7	962.5
G	162.5	6	975
Total		58	4750

$$Y_c = \frac{\sum D_y S_x}{n} = \frac{4750}{58},$$

$$Y_c = 81.896m$$

- Therefore, the centroid of the field is at $(X_c, Y_c) = (105.603m, 81.896m)$

Cont'd...

Example 2:

A topographic survey of a field gave the following elevations (m) at grid points. The field is required to be leveled to zero slope in all directions. The grid points are 25m apart in both directions. The reference line is at 12.5m from the first stake in both direction.

- a. Calculate the elevation of the centroid of the field.
- b. Calculate the cut and fill at the grid points.
- c. Compare the quantities of earthwork in cutting and filling.

	1	2	3	4	5
A	10.65	10.43	10.07	9.68	9.67
B	10.47	10.42	9.95	9.84	9.75
C	10.32	10.08	9.92	9.65	9.48
D	9.89	9.48	9.67	9.41	9.13

Cont'd...

○ Solution

- **Step 1:** Determining the average elevation of the field (elevation of the centroid)
 - Total number of stations = $4 * 5 = 20$
 - Total sum of elevations for all 20 stations = 197.96m

$$E_{avg} = \frac{\sum E_i}{n}$$

$$\sum E_i = 197.96m$$

$$n = 20$$

$$E_{avg} = \frac{197.96m}{20}$$

$$E_{avg} = \mathbf{9.898m}$$

Cont'd...

- Step 2: Determining the centroid of the field

Line	Distance, D_x	No stakes, S_y	Product
1	12.5	4	50
2	37.5	4	150
3	62.5	4	250
4	87.5	4	350
5	112.5	4	450
Total		20	1250

Line	Distance, D_y	No stakes, S_x	Product
A	12.5	5	62.5
B	37.5	5	187.5
C	62.5	5	312.5
D	87.5	5	437.5
Total		20	1000

$$X_c = \frac{\sum D_x S_y}{n} = \frac{1250 \text{ m}}{20}$$

$$X_c = 62.5 \text{ m}$$

$$Y_c = \frac{\sum D_y S_x}{n} = \frac{1000}{20},$$

$$Y_c = 50 \text{ m}$$

- Therefore, the centroid of the field is at $(X_c, Y_c) = (62.5 \text{ m}, 50 \text{ m})$

Cont'd...

- **Step 3:** Determine cut and fill at each grid points

The cuts and fills at various grid points are obtained by subtracting the elevation of the corresponding grid point from the elevation of the centroid.

	1	2	3	4	5
A	-0.752	-0.532	-0.172	+0.218	+0.228
B	-0.572	-0.522	-0.052	+0.058	+0.148
C	-0.422	-0.182	-0.022	+0.248	+0.418
D	+0.008	+0.418	+0.228	+0.488	+0.768

- Check the volume of cut and volume of fill. These two should balance each other.
- The –sign indicates the cut, while +sign indicates fill.

$$\sum Cut = \sum fill = 3.228\text{m}$$

End of Chapter

Thank You!

CHAPTER FIVE

Water Conveyance and Control Structure



CHAPTER FIVE

Water Conveyance and Control Structure

❖ Irrigation Distribution Systems

- The irrigation network consists of a **main intake structure** or **main pumping station**, a **conveyance system**, a **distribution system**, a **field application system**, and a **drainage system**.
- The **main intake structure**, or **main pumping station**, directs water from the source, such as a reservoir or a groundwater well or river, into the irrigation system.
- The **conveyance system** assures the transport of water from the main intake structure or main pumping station up to the field ditches.
- The **distribution system** assures the transport of water through field ditches to the irrigated fields.

Cont'd . . .

- The field application system assures the transport of water within the fields.
- The drainage system removes the excess water (caused by rainfall and/or irrigation) from the fields.

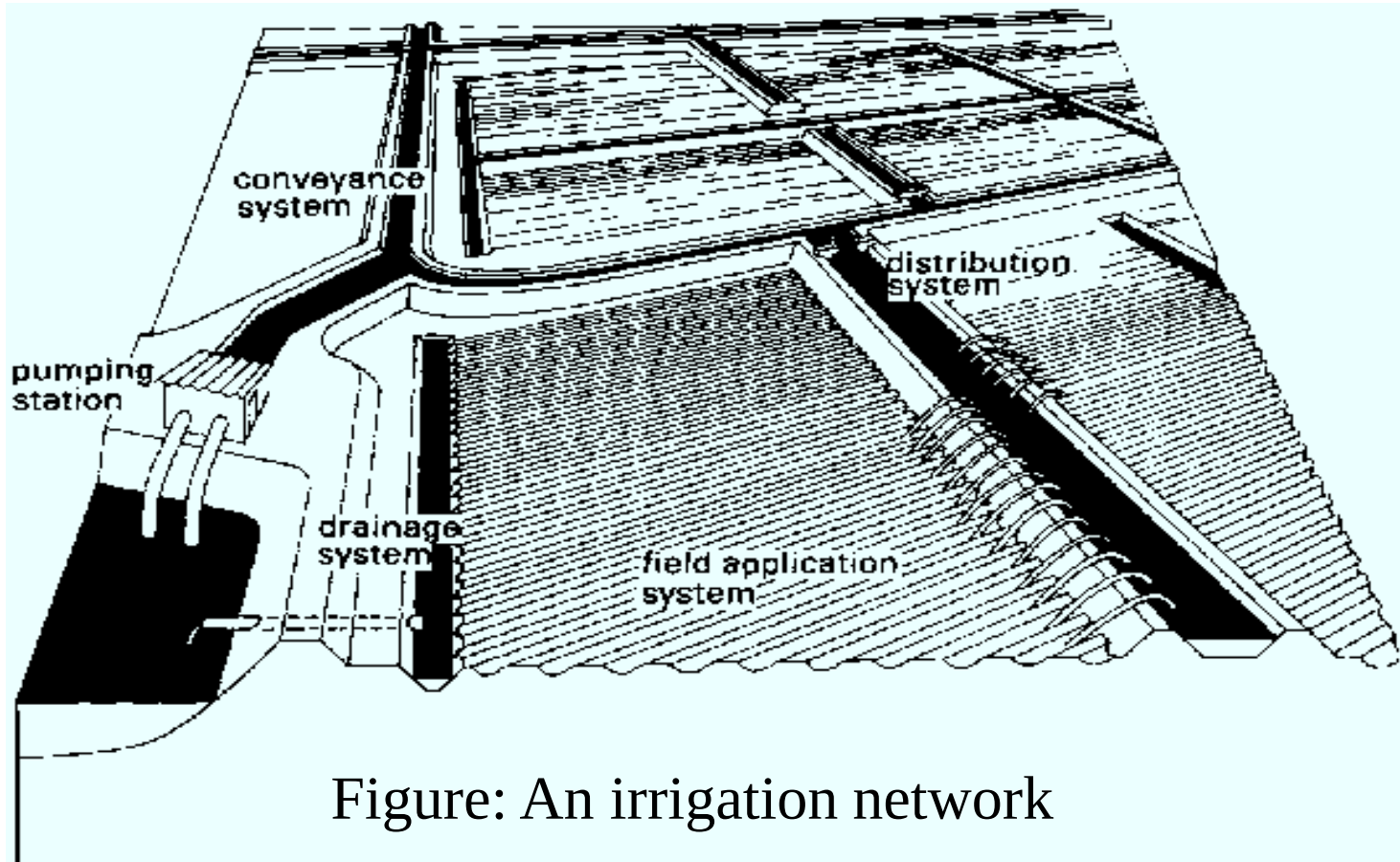


Figure: An irrigation network

Cont'd...

Main intake structure

- Intake structure is built at the entry to the irrigation system.
- Its purpose is to direct/divert water from the original source of supply (lake, river, reservoir etc.) into the irrigation system.



Cont'd...

Conveyance and distribution system

- The conveyance and distribution systems consist of canals transporting the water through the whole irrigation system.
- Canal structures are required for control and measurement of the water flow.
- Channels used for irrigation are normally called canals.
- Channels used for drainage are normally called drains.
- Field ditches have smaller dimensions and convey water from the farm entrance to the irrigated fields.



Cont'd...

Design of Irrigation Canal

○ Hydraulic radius (R)

- It is called also hydraulic mean depth.
- Is a length parameter that measures how efficiently a canal transmit water.
- The higher hydraulic radius means a larger cross-sectional area and greater flow capacity.
- It is the ratio of cross-sectional area to wetted perimeter

$$R = A/P_w$$

Best hydraulic section of channel

- Is the one that has the minimum or smallest wetted perimeter for a given discharge/area.
- It has also the following properties:
 - › The maximum flow capacity for the same flow cross-area
 - › The minimum cross-area and wetted perimeter for the same discharge
 - › The construction cost can be reduced nearly to the minimum

Cont'd...

Design of Irrigation Canal . . .

- The best hydraulic section for a trapezoidal canal is a half-hexagon.
- This means that the length of the sloping side of trapezoid should be equal to half of the top width. $R = y/2$.
- The best hydraulic section for rectangular canals is $b/y = 2$.

● Full supply discharge (FSD)

- It is the sum of the total irrigation requirement and the various losses in the conveyance of water
- The level of water in the canal at FSD is called full supply level (FSL).

Cont'd...

Design of Irrigation Canal . . .

○ Determination of canal design discharge

1. Determine the total area irrigated from the canal under, A ;
2. Based on the crop water requirement, determine the water demand for each unit irrigated in l/s/ha, q ;
3. For each outlet in the canal reach, determine the outflow by multiplying the area by the water requirement, $Q_i = Aq$;
4. Determine the outflow into the next reach of the canal, Q_o ;
5. Add all the outlet discharges and the outflow into the next reach of the canal to determine the design discharge of the canal reach under consideration and any losses.

The outlet discharges are;

$$Q_i = Aq \quad \text{where } A = \text{Area to be irrigated from the outlet}$$
$$q = \text{peak water demand rate (l/s/ha)}$$

The design discharge of the reach is thus,

$$Q = Q_1 + Q_2 + Q_3 + \dots + Q_o + \text{losses}$$

where: $Q_1, Q_2, Q_3, \dots$ are outlet discharges d/s of the reach and Q_o outflow into the next reach.

Cont'd...

Design of Irrigation Canal . . .

○ Critical velocity

- It is the velocity of water flowing in the canal when **no silting** or **no scouring** is taking place.
- It is recognized that sand deposits are not produced in a canal at a mean velocity of **0.45 - 0.9m/sec**.
- Similarly, water weeds hindering the flow capacity of the canal will not grow when the mean velocity is more than **0.7m/sec**.

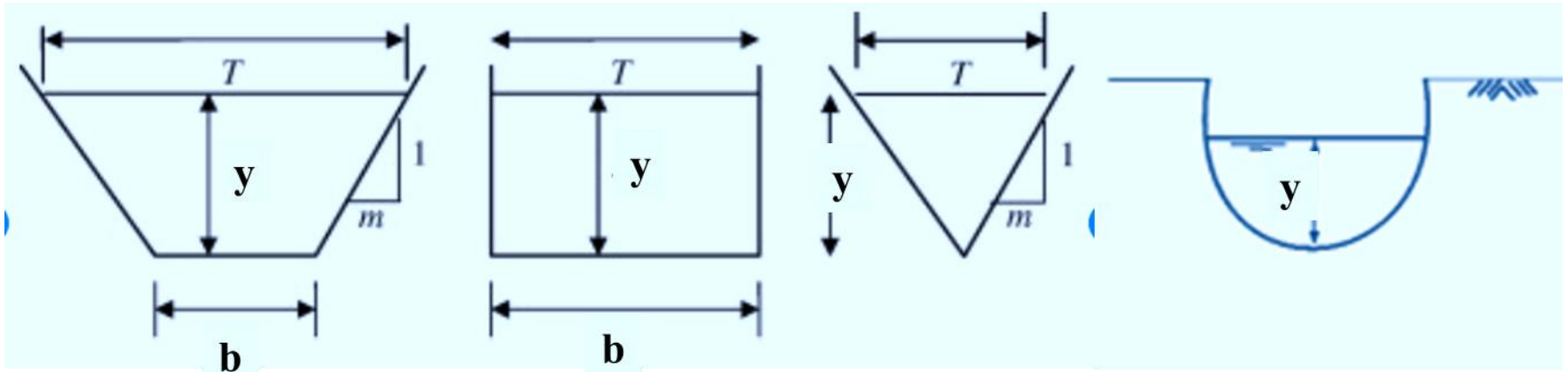
Maximum Allowable Velocity			
Type of material	Velocity (m/s)	Classification	Velocity (m/s)
Sandy soil	0.45	Thick concrete (approximately 18 cm)	3.00
Sandy loam	0.60	Thin concrete (approximately 10 cm)	1.50
Loam	0.70	Asphalt	1.00
Clayey loam	0.90	Block cavity wall (buttress pier less than 30 cm)	1.50
Clay	1.00	Block cavity wall (buttress pier 30 cm or larger)	2.00
Sandy clay	1.20	Block mortar masonry	2.50
Soft rock	2.00	Reinforced concrete pipe	3.00
Semi-hard rock	2.50	Steel pipe, ductile cast iron pipe	5.00
Hard rock	3.00	Petrochemical products group (polyvinyl chloride pipe, reinforced plastic composite tube)	5.00
		Reinforced concrete secondary product canal (excluding fence culvert)	3.00

Cont'd...

Design of Irrigation Canal . . .

Canal cross section

- The canal cross section can be rectangular, trapezoidal, triangular or circular.



Canal bed slope

- Large canals, $Q > 15 \text{ m}^3/\text{s}$,0.10 to 0.30 %
- Intermediate canals,0.20 to 0.40 %
- Small canals, $Q < 0.3 \text{ m}^3/\text{s}$,0.30 to 0.50 %

Cont'd...

□ Design of lined canals

- Here canal is covered (lined) with harder material like cement concrete, plastered stone masonry etc. to provide resistance against erosion, avoid seepage losses and growth of weeds.
- Materials which can be used for lining include cement concrete, plastered stone masonry, pre-cast concrete slabs, brick with sandwiched mortar and soil cement.
- The construction cost is much higher than for earthen canals.
- Maintenance is reduced for lined canals, but skilled labour is required.
- For the design of lined canals, Manning's or Chezy's equation can be used;

$$V = \frac{1}{n} * R^{\frac{2}{3}} * S^{\frac{1}{2}}$$

Manning's Equation

$$V = C\sqrt{RS}$$

Chezy's Equation

Cont'd...

□ Design of lined canals

- The following are recommended values of the Manning's roughness coefficient, n for different linings.
 - Concrete lining..... 0.013 to 0.018
 - Masonry lining with random stone..... 0.017 to 0.020
 - Brick lining..... 0.014 to 0.017
 - Asphalt lining..... 0.013 to 0.016

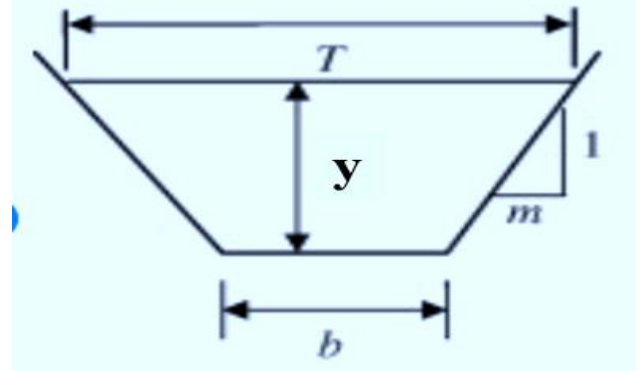


Cont'd...

□ Design of lined canals . . .

● Design of lined trapezoidal canal

$$A = by + my^2$$
$$P = b + 2y\sqrt{1 + m^2}$$
$$R = \frac{A}{P}$$
$$Q = A * V$$



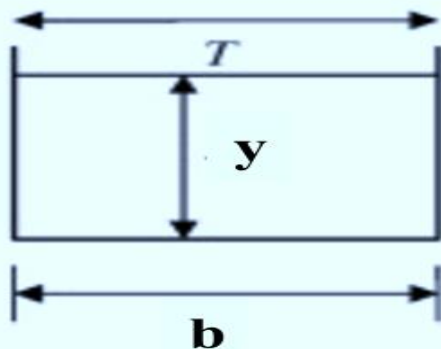
Where y is flow depth, b is bottom width, m is side slope, A is the area of flow and P is the wetted perimeter.

Continuity equation

● Velocity can be calculated using Manning's equation.

$$V = \frac{1}{n} * R^{\frac{2}{3}} * S^{\frac{1}{2}}$$

● Design of lined rectangular canal



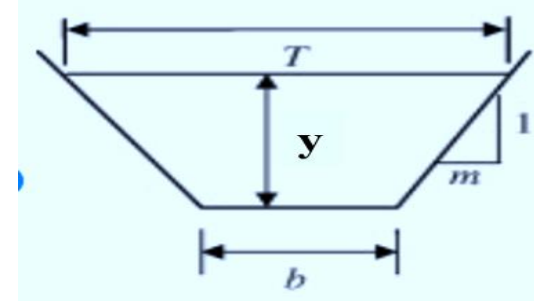
$$A = by$$
$$P = b + 2y$$
$$R = \frac{A}{P}$$
$$Q = A * V$$
$$V = \frac{1}{n} * R^{\frac{2}{3}} * S^{\frac{1}{2}}$$

Cont'd...

Example 1:

- Design a concrete lined trapezoidal canal which can supply a discharge of $Q = 0.3 \text{ m}^3/\text{s}$ to irrigate command area. Use a Manning's roughness coefficient, $n = 0.018$ (for concrete lining) and assume longitudinal slope, $S = 0.45\%$. Use $m = 1.5$. The permissible velocity should be between 0.7 m/s and 1.5 m/s .

Cont'd...



○ Solution

Step 1: From definition of hydraulic radius, express **b**, in terms of **y**.

$$A = by + 1.5y^2$$

$$P = b + 2y\sqrt{1 + 1.5^2}$$

$$R = \frac{A}{P} = \frac{by + 1.5y^2}{b + 2y\sqrt{1 + 1.5^2}} \quad \dots\dots\dots 1$$

From efficient and economic point of view, $R = y/2$ 2

Hence, equate equation 1 and 2;

$$\frac{by + 1.5y^2}{b + 2y\sqrt{1 + 1.5^2}} = \frac{y}{2}$$

Solving this equation yields:

$$\mathbf{b = 0.606y}$$

Cont'd...

- Step 2: From Manning's equation, calculate **b** and **y**.

$$Q = \frac{1}{n} * A * R^{\frac{2}{3}} * S^{\frac{1}{2}}$$

$$0.3 = \frac{1}{0.018} * (by + 1.5y^2) * \left(\frac{by + 1.5y^2}{b + 3.606y} \right)^{\frac{2}{3}} * 0.0045^{\frac{1}{2}}$$

- Substitute the value of **b=0.606y** into the above equation and find the values of **b** and **y**

$$0.3 = \frac{1}{0.018} * (0.606y * y + 1.5y^2) * \left(\frac{0.606y * y + 1.5y^2}{0.606y + 3.606y} \right)^{\frac{2}{3}} * 0.0045^{\frac{1}{2}}$$

- Solving the above equation yields:

$$\mathbf{y = 0.35m}$$

$$B = 0.606*y \rightarrow b = 0.606*0.35m \rightarrow \mathbf{b=0.212m}$$

Cont'd...

- **Step 3:** calculate the flow velocity in the canal and compare it with the allowable velocity.

$$A = by + my^2$$

$$A = 0.212 * 0.35 + 1.5 * 0.35^2$$

$$A = \mathbf{0.258m^2}$$

$$P = b + 2y\sqrt{1 + m^2}$$

$$P = 0.212 + 2 * 0.35\sqrt{1 + 1.5^2}$$

$$P = \mathbf{1.474m}$$

$$R = \frac{A}{P} = \frac{0.258m^2}{1.474m}$$

$$R = \mathbf{0.175m}$$

Cont'd...

$$V = \frac{1}{n} * R^{\frac{2}{3}} * S^{\frac{1}{2}}$$

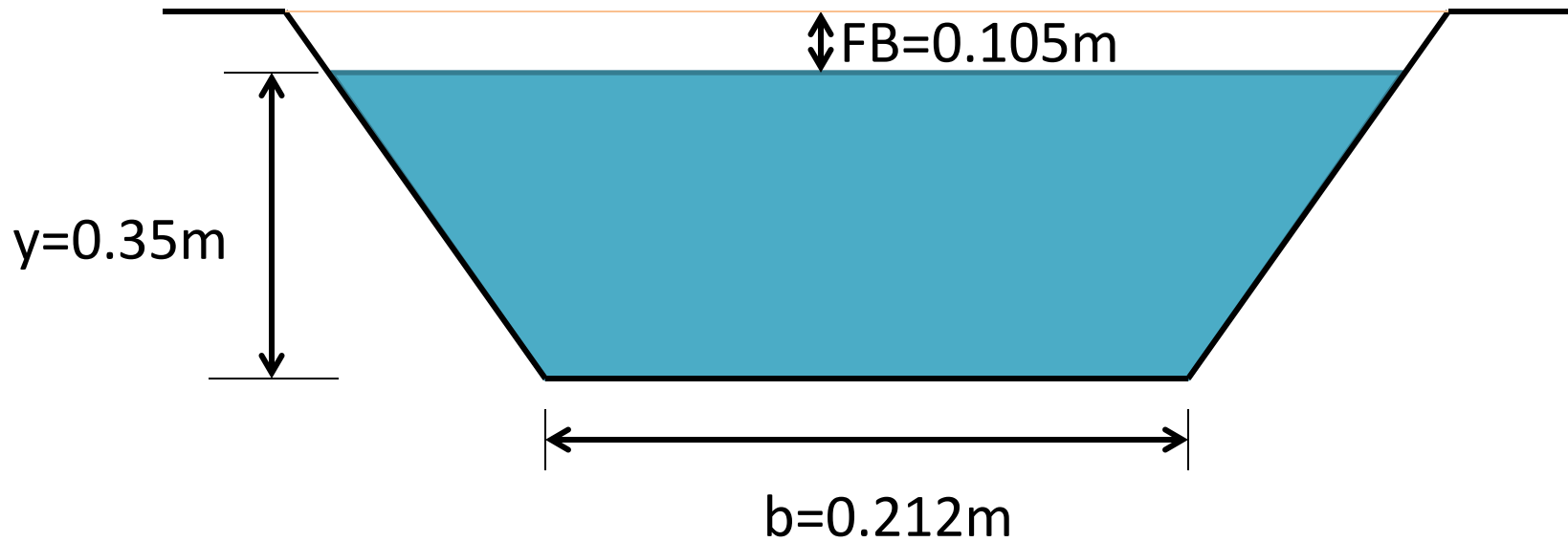
$$V = \frac{1}{0.018} * 0.175^{\frac{2}{3}} * 0.0045^{\frac{1}{2}}$$

$$V = 55.556 * 0.31285 * 0.06708$$

$$V = 1.166m/s$$

- The minimum flow velocity that will not allow sediment deposition in the canal is **0.7m/s**.
- Similarly, the maximum flow velocity that will not erode canal bed is **1.5m/s** for thin concrete lined canal (see table 3.1).
- The calculated velocity, **V=1.166m/s** is within the allowable range.
- Hence the designed canal dimensions are acceptable.

Cont'd...



Cross-section of the designed canal

Cont'd...

□ Design of Unlined canals

- Unlined canals can be classified into two classes based on the stability of the boundaries of the canal for design purposes:
 - A. Canals with stable (non-erodible) bed and
 - B. Canals with erodible bed (Alluvial)

A. Design of non-erodible (stable) canals:

- Non-erodible canals are canals with fairly stable boundary.
- The design of such canals should ensure that any sediment entering into the canal go on flowing and avoid settlement.
- For design of such canals, equations employed:
 - I. Continuity equation;
 - II. Manning formula
- There are two approaches for the design of such canals:
 - I. The recommended (b/d ratio) approach and
 - II. The Tractive force (permissible velocity) approach.

Cont'd...

□ Design of Unlined canals . . .

A. Design of non-erodible (stable) canals . . .

I. **b/d ratio approach**

- The **best b/d ratio** is given by

$$b/d = 2[(1 + z^2)^{0.5} - z]$$

- Recommended side slopes that are stable under normal conditions are as follows:

Material	z
Rock	0
Stiff clay	0.5
Cohesive medium soils	1.0 - 1.5
Sand	2
Fine sand, porous clay, soft peat	3

Cont'd...

□ Design of Unlined canals . . .

A. Design of non-erodible (stable) canals . . .

I. b/d ratio approach . . .

- Recommended values for z and b/d for unlined irrigation canals constructed in earth (cohesive medium soils) as related to the design discharge of the canal Q are as follows:

Q (m ³ /s)	0.2	0.2 – 0.5	0.5 – 10.0	>10.0
z	0.8 - 1	1 – 1.5	1.5 - 2	> 2
b/d	1	$1.76 * Q^{0.35}$		

- In most small and medium size earth canals, the freeboard varies from 50% to 60% of the depth y , with a minimum of 0.15 or 0.20 m.

Cont'd...

Example 2:

Design the section of an unlined canal in a loamy soil to carry a discharge of 50 cubic meter per second with permissible velocity of one meter per second. Assume side slope 2:1 and B/D ratio as 6. Using Manning formula calculate bed slope of the canal. Take $n = 0.0225$.

Cont'd...

Solution:

Using fundamental equation $Q = AV$

$$A = Q/V = 50/1 = 50 \text{ m}^2$$

$$A = by + my^2 \Rightarrow 50 = (b + 2y)y$$

$$P_w = b + 2y\sqrt{1 + m^2} = b + 2\sqrt{5}y$$

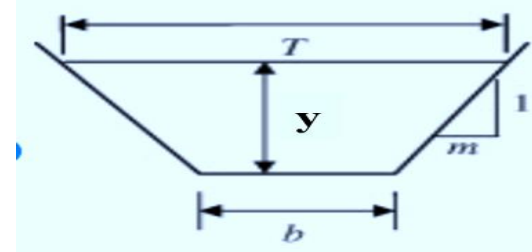
$$b/y = 6 \Rightarrow b = 6y$$

Substituting b in to A

$$50 = (6y + 2y)y = 8y^2 \Rightarrow y = \sqrt{\left(\frac{50}{8}\right)}$$

$$\Rightarrow D = 2.5\text{m}$$

$$B = 6D = 6 \times 2.5 = 15\text{m}$$



$$\begin{aligned} A &= by + my^2 \\ P &= b + 2y\sqrt{1 + m^2} \\ R &= \frac{A}{P} \\ Q &= A * V \end{aligned}$$

Cont'd...

Slope from Manning formula

$$V = \frac{1}{n} . R^{\frac{2}{3}} . S^{\frac{1}{2}}$$

Hydraulic radius

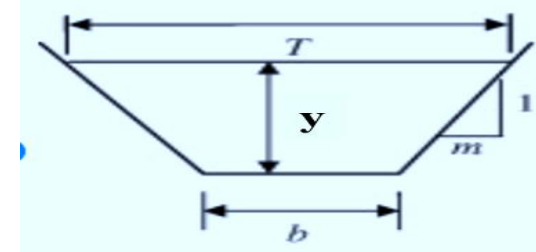
$$R = \frac{A}{P_w} = \frac{50}{15 + 2 * \sqrt{5} * 2.5} = \frac{50}{26.18} = 1.91m$$

Substitute the value of R in to V

$$1 = \frac{1}{0.0225} * (1.91)^{\frac{2}{3}} * S^{\frac{1}{2}} = \frac{1.5}{0.0225} 4 * S^{\frac{1}{2}}$$

$$S = \left(\frac{0.0225}{1.54} \right)^2$$

$$S = \frac{1}{4740}$$



$$\begin{aligned} A &= by + my^2 \\ P &= b + 2y\sqrt{1 + m^2} \\ R &= \frac{A}{P} \\ Q &= A * V \end{aligned}$$

Cont'd...

❑ Design of Unlined canals . . .

A. Design of non-erodible (stable) canals . . .

II. On Permissible velocity approach.

- *A minimum permissible velocity* is that which **will not start sedimentation** and *maximum permissible velocity* is that w/c **will not cause erosion of the canal**.

Table: **Maximum permissible velocities** and the corresponding unit-tractive force values

Material	Clear water		Loaded water	
	Velocity, m/s	Tc (N/m ²)	Velocity, m/s	Tc (N/m ²)
Fine sand	0.46	1.3	0.76	3.61
Sandy loam	0.53	1.78	0.76	3.61
Silt loam	0.61	2.31	0.91	5.29
Alluvial silts	0.61	2.31	1.07	7.22
Volcanic ash	0.76	3.61	1.07	7.22
Stiff clay	1.14	12.51	1.52	22.13
Alluvial silts	1.14	12.51	1.52	22.13

Cont'd...

□ Design of Unlined canals . . .

B. Design of erodible (Alluvial) canals:

- Erodible canals are canals with movable bed.
- Canal design on erodible (alluvial deposits) not only have erosion problem but also water carries sediments with it.
- So, the design is made based on the **maximum** and **minimum permissible velocities**.
- This velocity w/c doesn't cause theoretically silt in the canals & erosion of the be is called **non-silting** & **non-scouring velocity**.
- Canals designed with this velocity are called **regime canals**.
- .A channel is in **state of regime means** that whatever sediment entering the canal at the head is kept in suspension so that it will not settle and local sediments are not produced by erosion of the canal beds

Cont'd...

□ Design of Unlined canals . . .

B. Design of erodible (Alluvial) canals . . .

- There are two theories for design of stable (non-silting and non-scouring) alluvial canals.
 - Kennedy's Theory
 - Lacey Theory

❖ Kennedy's Theory

- According to Kennedy, a critical velocity which will just keep the canal free from silting and scouring.

$$V_o = 0.546 m y^{0.64} \quad \text{or} \quad V_o = 0.546 m D^{0.64}$$

Where, V_o is critical velocity (m/s)

m is critical velocity ratio depends on the size of the silts. ($=V/V_o$)

y or D or d is the depth of flow or is Full supply depth (m)

0.55 and 0.64 are constants which depend on characteristics and type of silt.

Cont'd...

❑ B. Design of erodible (Alluvial) canals . . .

❖ Kennedy's Theory . . . Procedure

- Procedure for design of regime canals on Kennedy's theory
 1. Assume a trial depth of flow y and determine the critical velocity V_o ;
 2. Determine the area of flow, A from $A = Q/V_o$;
 3. Workout the canal cross sectional parameters (B , R ...);
 4. Calculate the actual mean velocity V in the canal from the Kutter's formula, Manning's formula or Chezy's equation;
 5. Compare V and V_o , If the same, the assumed depth of flow d is right if not the same assume another y and repeat steps 1 to 4.

Kutter's Formula

$$V = \left[\frac{\frac{1}{n} + \left(23 + \frac{0.00155}{S} \right)}{1 + \left(23 + \frac{0.00155}{S} \right) * \frac{n}{\sqrt{R}}} \right] \sqrt{RS}$$

Chezy's equation

$$V = C\sqrt{RS}$$

Where: V is the flow velocity, m/s R is Hydraulic radius, m
 S is Slope of the canal n is roughness coefficient
 C is Lacey's constant

Cont'd...

Example 3:

Design an irrigation canal for the following data:

Use Kennedy's equations FS discharge = $14 \text{ m}^3/\text{s}$

Bed slope = 1 in 5000 Kutter's $n = 0.0225$

Side slope = $\frac{1}{2} : 1$ (H : V) Assume CVR, V/V_0 (m) = 1

Solution:

1. Assume full supply depth, $y = 1.78 \text{ m}$

$$V_0 = 0.55 \text{ m } y^{0.64} = 0.55 * 1 * 1.78^{0.64} = 0.795 \text{ m/s}$$

2. $A = Q/V = 14/0.795 = 17.6 \text{ m}^2$

3. Calculate canal dimensions

$$A = by + my^2 \quad \Rightarrow \quad 17.6 = 1.78b + 0.5(1.78)^2$$

$$\therefore b = 9.0 \text{ m}$$

$$P_w = b + 2y\sqrt{1 + m^2} = 9.0 + 2(1.78)\sqrt{1 + 0.5^2} = 12.98 \text{ m}$$

$$R = \frac{A}{P_w} = \frac{17.6}{12.98} = 1.36 \text{ m}$$

Cont'd...

4. Using Chezy's calculate actual velocity:

Chezy's constant can be calculated from Kutter's formula:

$$C = \left[\frac{\frac{1}{n} + \left(23 + \frac{0.00155}{s} \right)}{1 + \left(23 + \frac{0.00155}{s} \right)} \right] = \left[\frac{\frac{1}{0.0225} + \left(23 + \frac{0.00155}{0.0002} \right)}{1 + \left(23 + \frac{0.00155}{0.0002} \right)} \right] = 46.8$$

$$V = C\sqrt{RS} = 46.8 \sqrt{1.36 \times 0.0002} = 0.772 \text{ m/s} < 0.795 \text{ m/s}$$

Thus, this velocity come out to be slightly less than velocity obtained from Kennedy's equation (critical velocity). Hence one more trials with smaller depth will give fairly closes result.

Therefore, take $y = 1.712\text{m}$ and $b = 9.76\text{m}$

Cont'd...

Example 4:

Assume, Full supply depth, $D = 1.25$ m

Critical velocity, $V_o = 0.546 \times 1 \times 1.25^{0.64} = 0.629$ [Assume, $m = 1$]

As $m = 1$, $V = V_o$

$$\therefore A = 6/0.629 = 9.53 \text{ m}$$

$$A = 1.25B + 1.56 \Rightarrow B = 6.38 \text{ m}$$

$$P = B + 2\sqrt{2} \times 1.25 = 9.92 \text{ m}$$

$$R = 0.96 \text{ m}$$

$$\text{By Kutter's formula, } C = \left[\frac{\frac{1}{0.0225} + \left(23 + \frac{0.00155}{0.0002} \right)}{1 + \left(23 + \frac{0.00155}{0.0002} \right) \frac{0.0225}{\sqrt{0.96}}} \right] = 44.23$$

By Chezy's formula, $V = 44.23 \times \sqrt{(1 \times 0.0002)} = 0.613 \text{ m/sec}$

$$C.V.R = 0.613/0.629 = 0.97 < 1$$

In this case, the CVR is very close to 1. So, the design may be accepted. So, finally,

$$D = 1.25 \text{ m and } B = 6.28 \text{ m}$$

Cont'd...

❑ B. Design of erodible (Alluvial) canals . . .

❖ Lacey's Regime Theory

- He introduced a silt factor (f) which is a function of the mean particle size
- Lacey also states that the silt is kept in suspension solely by force of eddies. But also, Lacey adds that eddies are not only generated on the bed but at all points on the wetted perimeter.
- Unlike Kennedy, Lacey takes hydraulic radius R as a variable rather than depth y or d .
- Lacey also argued that bed material size is an important parameter to be considered.
- As a result, he introduced a silt factor (f) which is a function of the mean particle size.

Cont'd...

❑ B. Design of erodible (Alluvial) canals . . .

❖ Lacey's Regime Theory . . . Procedure

- Procedure for design of regime canals on Lacey's Theory

1. Evaluate the flow velocity from $V = 0.4382(Qf^2)^{1/6}$

where V is in m/s

Q is design discharge in m³/s

f is silt factor, $f = 1.76 * \sqrt{d_m}$, where d_m is mean particle size, mm

2. Determine the hydraulic radius, R from: $R = 2.46 \left(\frac{V^2}{f} \right)$

3. Calculate the area of flow, A $A = Q/V$

4. Calculate wetted perimeter, P $P = 4.75 * \sqrt{Q}$

5. Workout y and b from the known, A, P and R.

6. Compute the canal bed slope, S $S = \left[\frac{f^{5/3}}{3316 Q^{1/6}} \right]$

Cont'd...

Example 5:

Design an irrigation canal Using Lacey's theory from the following data:

$$\text{FS discharge} = 14\text{m}^3/\text{s} \quad f = 1$$

$$\text{Side slope} = \frac{1}{2} : 1 \text{ (H : V)}$$

$$\text{Coefficient of rugosity, } n = 0.0225$$

Assume any other data required suitably.

Solution:

1. Calculate mean velocity

$$V = 0.4382(Qf^2)^{1/6} = 0.43885 (14 * 1^2)^{1/6} = 0.681 \text{ m/s}$$

2. Determine the hydraulic radius, R from

$$R = 2.46 * \frac{V^2}{f} = 2.46 * \frac{0.68^2}{1} = 1.14\text{m}$$

3. Calculate the area of flow

$$A = Q/V = 14/0.681 = 20.55 \text{ m}^2$$

Cont'd...

4. Calculate wetted perimeter, P_w

$$P_w = 4.75 * \sqrt{Q} = 4.75 * \sqrt{14} = 17.77\text{m}$$

5. Workout **y** and **b** from the known, A, P and R.

$$- A = by + my^2 \quad \Rightarrow \quad 20.55 = by + 0.5y^2$$

$$- P_w = b + 2y\sqrt{1 + m^2} \quad \Rightarrow \quad 17.77 = b + \sqrt{5} y$$

$$\text{Hence, } b = 17.77 - \sqrt{5} y$$

- Substitute values of b in equation of A

$$20.55 = (17.77 - \sqrt{5} y)y + 0.5y^2$$

$$20.55 = 17.77y - 1.73y^2$$

$$\mathbf{1.73y^2 - 17.77y + 20.55 = 0}$$

- By using quadratic equation

$$y = \frac{+17.77 \pm \sqrt{17.77^2 - 4 * 1.73 \times 20.55}}{2 \times 1.73}$$

$$y = \frac{17.77 \pm 13.175}{3.46} = 8.94 \text{ \& } 1.32\text{m}$$

Cont'd...

- Taking $y = 8.94\text{m}$, $b = 17.77 - 8.94\sqrt{5} = -2.22$ is illogical
- Taking $y = 1.3\text{ m}$, $b = 17.77 - 1.32\sqrt{5} = 14.82\text{m}$
Hence take **$y = 1.32\text{ m}$** and **$b = 14.82\text{ m}$**
- The value of y & b can be utilized to calculate hydraulic radius and compare it with already calculated from Lacey.

$$R = \frac{A}{P_w} = \frac{14.82 \times 1.32 + 0.5 \times 1.32^2}{14.82 + 1.32\sqrt{5}} = \frac{20.43}{17.77} = 1.14\text{m}$$

6. Compute the canal bed slope, S

$$S = \left[\frac{f^{5/3}}{3316 \cdot Q^{1/6}} \right] = \frac{1^{5/3}}{3316 * (14)^{1/6}} = \frac{1}{3316 * 1.553} = \frac{1}{5160}$$

$\therefore$ F.S depth 1.32m, bed width 14.82 m and bed slope 1 in 5160.

Cont'd...

Example 6:

Design an irrigation canal Using Lacey's theory when following data is given:

Mean diameter of silt particles = 0.33mm

Longitudinal slope of channel = 1 in 6000

Side slope = $\frac{1}{2} : 1$ (Horizontal : Vertical)

Coefficient of rugosity, $n=0.0225$

Assume that for the given slope maximum discharge the canal has to carry under regime conditions.

Example 5: Design an irrigation canal Using Lacey's theory when following data is given:

Mean diameter of silt particles = 0.33mm

Longitudinal slope of channel = 1 in 6000

Side slope = $\frac{1}{2} : 1$ (Horizontal : Vertical)

Coefficient of rugosity, $n = 0.0225$

Assume that for the given slope maximum discharge the canal has to carry under regime conditions.

Solution:

Step 1 Silt factor

$$\begin{aligned} f &= 1.76 * \sqrt{d_m} \\ &= 1.76 * \sqrt{0.33} = 1.0 \end{aligned}$$

Step 2 Determine design discharge

$$S = \left[\frac{f^{\frac{5}{3}}}{3316 * Q^{\frac{1}{6}}} \right]$$

$$Q = \left[\frac{f^{\frac{5}{3}}}{3316 * S} \right]^6 = 35.1$$

Step 3 Mean velocity

$$V = 0.4382(Qf^2)^{\frac{1}{6}} = 0.4382 * (35.1 * 1^2)^{\frac{1}{6}} = 0.793m / s$$

Step 4 Determine hydraulic radius

$$R = 2.46 * \frac{V^2}{f} = 2.46 * \frac{0.793^2}{1} = 1.54m$$

$$P_w = 4.825 * \sqrt{Q} = 4.825 * \sqrt{35.1} = 28.59m$$

Step 4 Cross sectional area

$$A = R * P_w$$

$$= 1.54m * 28.59m = 44.05m^2$$

$$A = BD + 0.5D^2$$

$$44.05 = BD + 0.5D^2$$

$$P_w = B + D\sqrt{5}$$

$$28.59 = B + D\sqrt{5} \Rightarrow B = 28.59 - 2.23D$$

$$44.05 = (28.59 - 2.23D)D + 0.5D^2 = 28.59D - 1.73D^2$$

$$1.73D^2 - 28.59D + 44.02 = 0$$

$$D = \frac{+ 28.59 \pm \sqrt{28.59^2 - 4 * 1.73 * 20.55}}{2 * 1.73} =$$
$$\frac{28.59 \pm 22.64}{3.46} = 14.8m \text{ \& } 1.72m$$

Taking $D=14.80\text{m}$, $B=28.59 - 2.23*14.8 = -4.5$ is absurd

Taking $D=1.72\text{m}$, $B=28.59 - 2.23*1.72 = 24.75\text{m}$

Hence taking $D=1.72\text{m}$ and $B = 24.75\text{m}$

❖ These value of D & B can be utilize to calculate value of R .

$$R = \frac{A}{P_w} = \frac{24.75*1.72 + (1.72)^2}{24.75 + \sqrt{5*1.72}} = 1.54\text{m}$$

Thus values of R is equal to the value calculated in step 4 above.

Example 6

Discharge, $Q=0.3\text{m}^3/\text{sec}$, Manning's roughness coefficient= 0.017 (for masonry lining), Longitudinal slope, $S=1\text{m}/1\text{km}$ and Canal cross-section is rectangular

Let assume $b/y=1.5$ i.e, $b=1.5y$

Flow in the canal expressed the equation

$$Q = \frac{1}{N} S^{1/2} A R^{2/3}$$

$$\frac{Q * N}{S^{1/2}} = A R^{2/3}$$

Wetted area wetted, $A_w = b * y = 1.5y * y = 1.5y^2$

Wetted peri meter, $P_w = b + 2y = 1.5y + 2y = 3.5y$

$$\text{Hydraulic readiness } R_w = \frac{A}{P} = \frac{1.5y^2}{3.5y} = 0.429y$$

$$\frac{0.30 * 0.017}{\sqrt{0.001}} = 1.5y^2 [0.429y]^{2/3}$$

$$0.1898 = y^{8/3}$$

$$Y = 0.536\text{m}$$

$$b = 0.8044\text{m take } b = 0.80\text{m}$$

Therefore adopt

Bed width, $b = 0.80\text{m}$

Total depth, $FB + y = 1.0\text{m}$

Design best hydraulic section rectangular canal use above data (HW)

Methods of Water Measurement

- The principal objective of measuring irrigation water is to permit efficient distribution and application.
- The most commonly used water measuring structures are
 - **Weirs or Barrages** and
 - **Flumes**
- In these structures, the water depth is read on a scale which is part of the structure.
- Using this reading, the flow-rate is then computed from standard formulas or obtained from standard tables prepared specially for the structure.

Cont'd...

❖ Weir or Barrage

Weir

- Is an impervious barrier which is constructed across a river to **raise the water level** on the upstream side section.
- Here the water level is raised up to the required height and the surplus water is allowed to flow over the weir.
- Generally, it is constructed across an inundation river.



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❖ Weir or Barrage . . .

Barrage

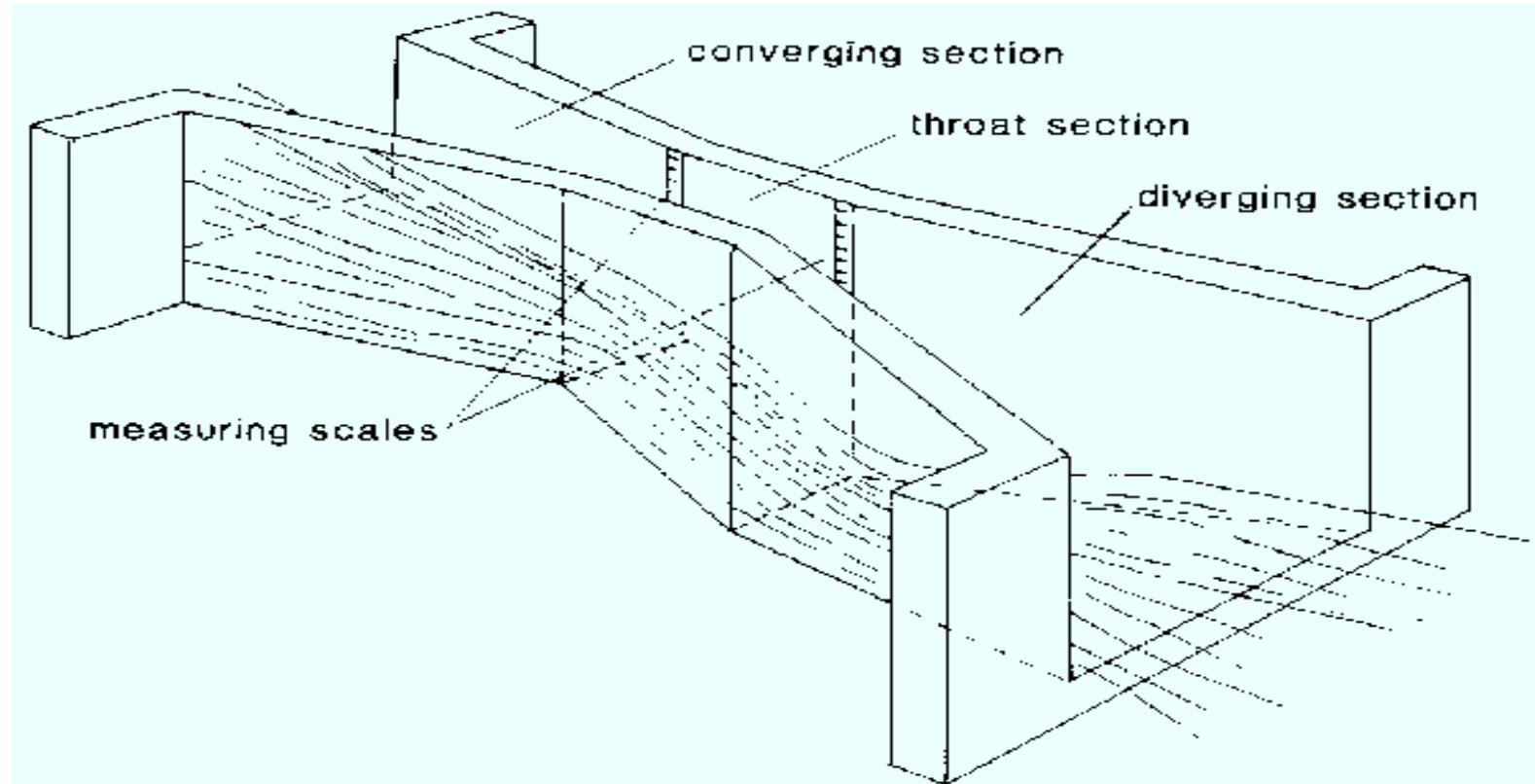
- When adjustable gates are installed over a weir to maintain the water surface at different levels at different times is known as a barrage.
- The water level is adjusted by operating the gates or shutters. The gates are placed at different tiers and these are operated by cables from the cabin.
- The gates are supported on piers at both ends. The distance between the pier to pier is known as Bay.



Cont'd...

❖ Parshall Flumes

- The Parshall flume consists of a metal or concrete channel structure with three main sections: (1) a **converging section** at the upstream end, (2) a **constricted or throat section** and (3) a **diverging section** at the downstream end



Cont'd...

❖ **Parshall Flumes . . .**



Cont'd...

❖ Long-throat flume

- The unique characteristic of these structures as compared to most other open-channel measurement devices is the fact that they **can be rated (calibrated) by computer analysis**, eliminating the need for laboratory calibration of each different flume or weir configuration.
- Basic components of long-throated flume flow measurement structures include an **approach channel** and **head-measurement location**, a **converging transition**, a **control section**, an **optional diverging transition**, and the **tail water channel**.

❖ Cut-throat flume

- The cut-throat flume is similar to the Parshall flume, but has no throat section, only converging and diverging sections.
- Unlike the Parshall flume, the cut-throat flume has a flat bottom.
- It is easier to construct and install, the cut-throat flume is often preferred to the Parshall flume.

Cont'd...

❖ Long-throat flume . . .

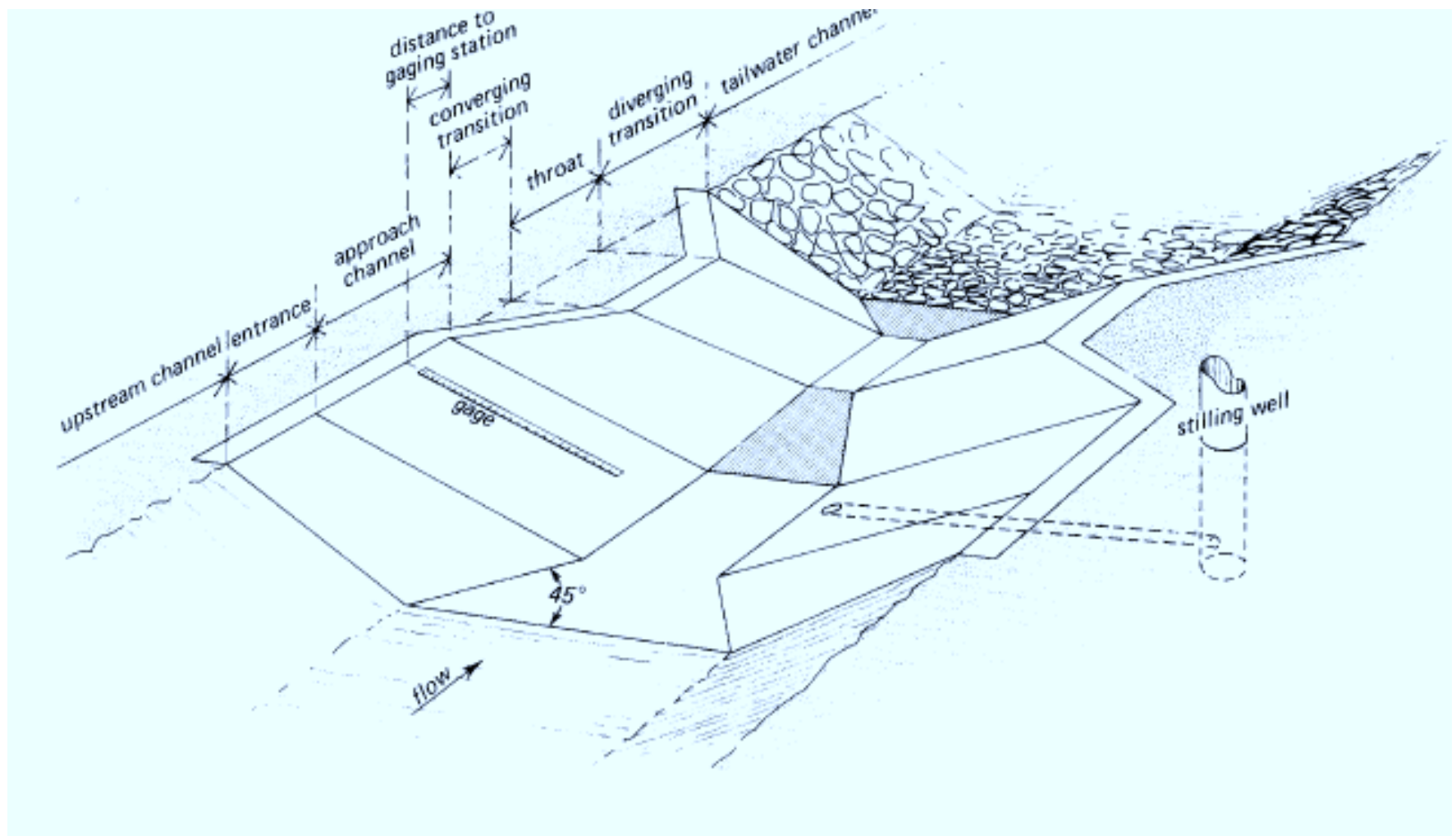


Figure: Long-throat flume

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❖ **Cut-throat flume . . .**

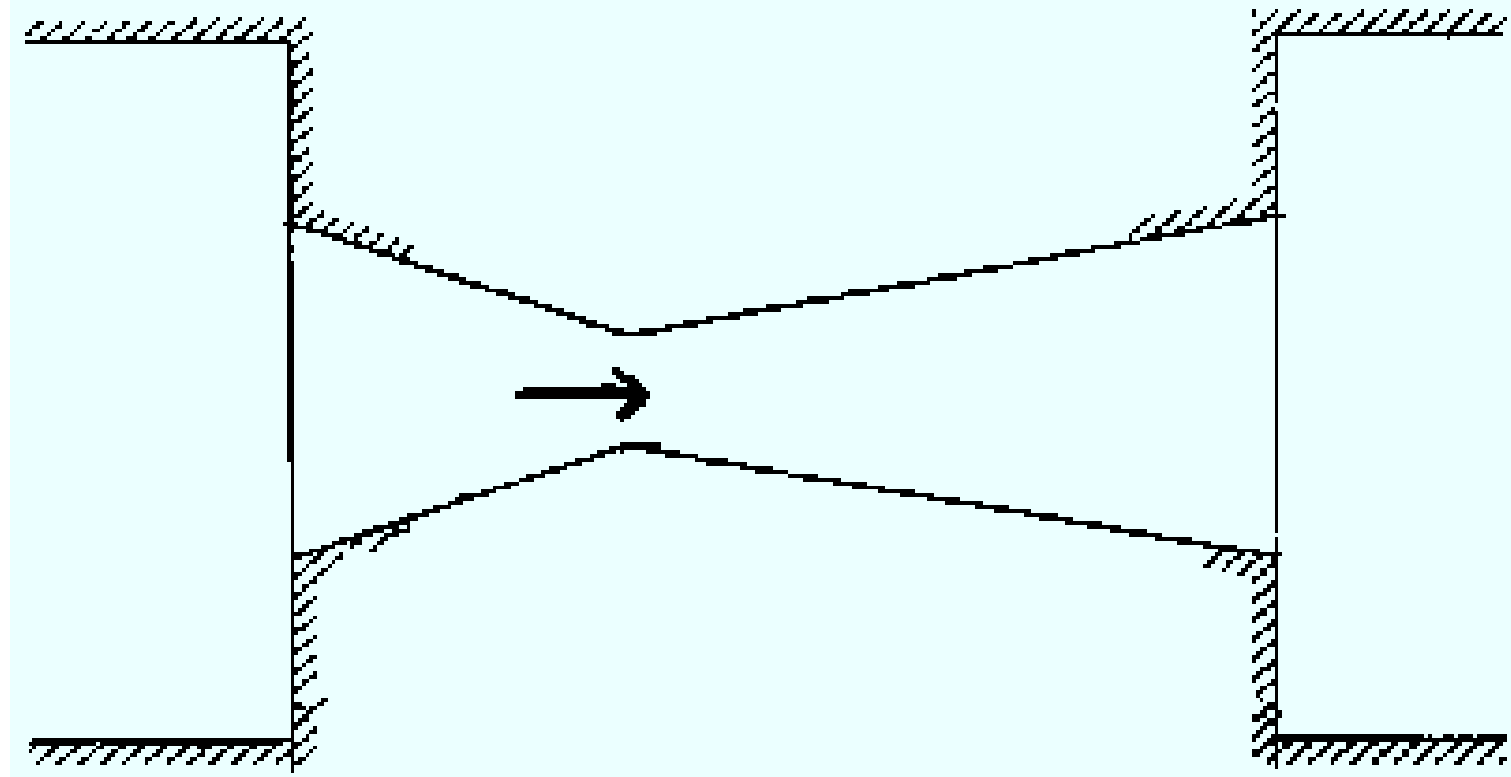


Figure: Cut-throat flume

Miscellaneous Irrigation Structures

- The flow of irrigation water in the canals must always be under control.
- For this purpose, canal structures are required. They help to regulate the flow and deliver the correct amount of water to the different branches of the system and onward to the irrigated fields.
- There are four main types of structures:
 - Erosion control structures,
 - Distribution control structures,
 - Crossing structures and
 - Water measurement structures.

Cont'd...

i. Erosion Control Structures

a) Canal erosion

- Water flowing in steep canals can reach very high velocities.
- Soil particles along the bottom and banks of an earthen canal are then lifted, carried away by the water flow, and deposited downstream where they may block the canal and silt up structures.
- The canal is said to be under erosion; the banks might eventually collapse.

b) Drop structures and chutes

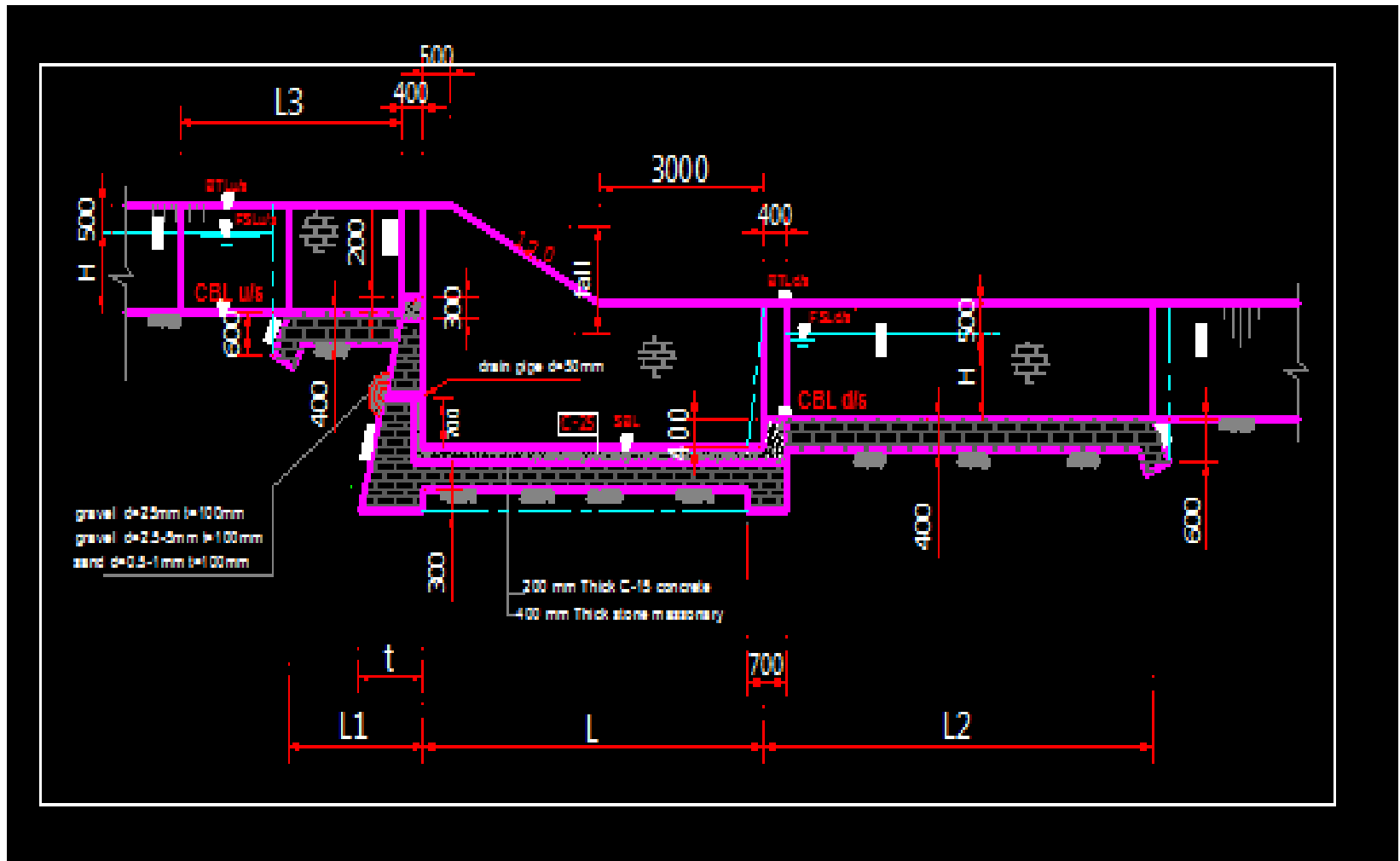
- Are required to reduce the bottom slope of canals lying on steeply sloping land in order to avoid high velocity of the flow and risk of erosion.
- These structures permit the canal to be constructed as a series of relatively flat sections, each at a different elevation.
- Drop structures take the water abruptly from a higher section of the canal to a lower one.
- In a chute, the water does not drop freely but is carried through a steep, lined canal section.
- Chutes are used where there are big differences in the elevation of the canal.

Cont'd...



Fig. Drop structures

Cont'd...



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ii. Distribution Control Structures

- Distribution control structures are required for easy and accurate water distribution within the irrigation system and on the farm.

a) Division boxes

- Division boxes are used to divide or direct the flow of water between two or more canals or ditches.
- Water enters the box through an opening on one side and flows out through openings on the other sides. These openings are equipped with gates.

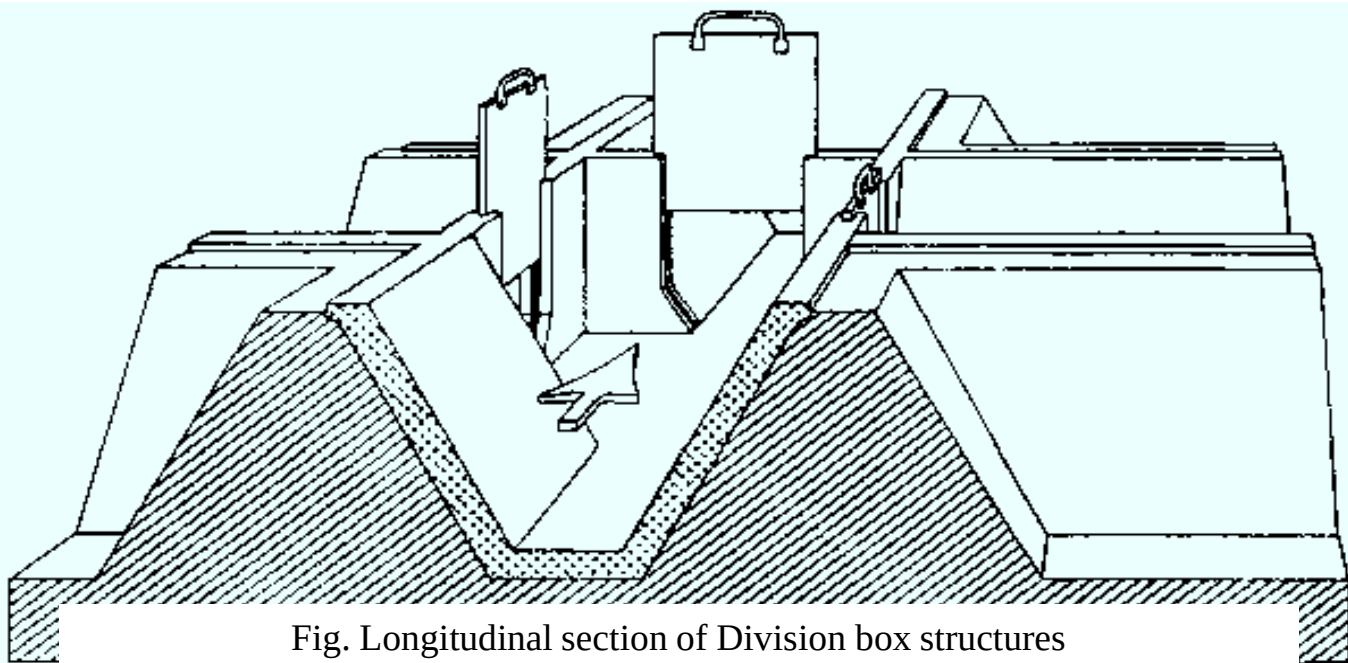


Fig. Longitudinal section of Division box structures

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Fig. Division box structures

Cont'd...

ii. Distribution Control Structures . . .

b) Turnouts

- Turnouts are constructed in the bank of a canal. They divert part of the water from the canal to a smaller one.
- Turnouts can be concrete structures or pipe structures.



Figure: A concrete turnout

Cont'd...

ii. Distribution Control Structures . . .

c) Checks

- Diverts water from the field ditch to the field, it is often necessary to raise the water level in the ditch.
- Checks are structures placed across the ditch to block it temporarily and to raise the upstream water level.

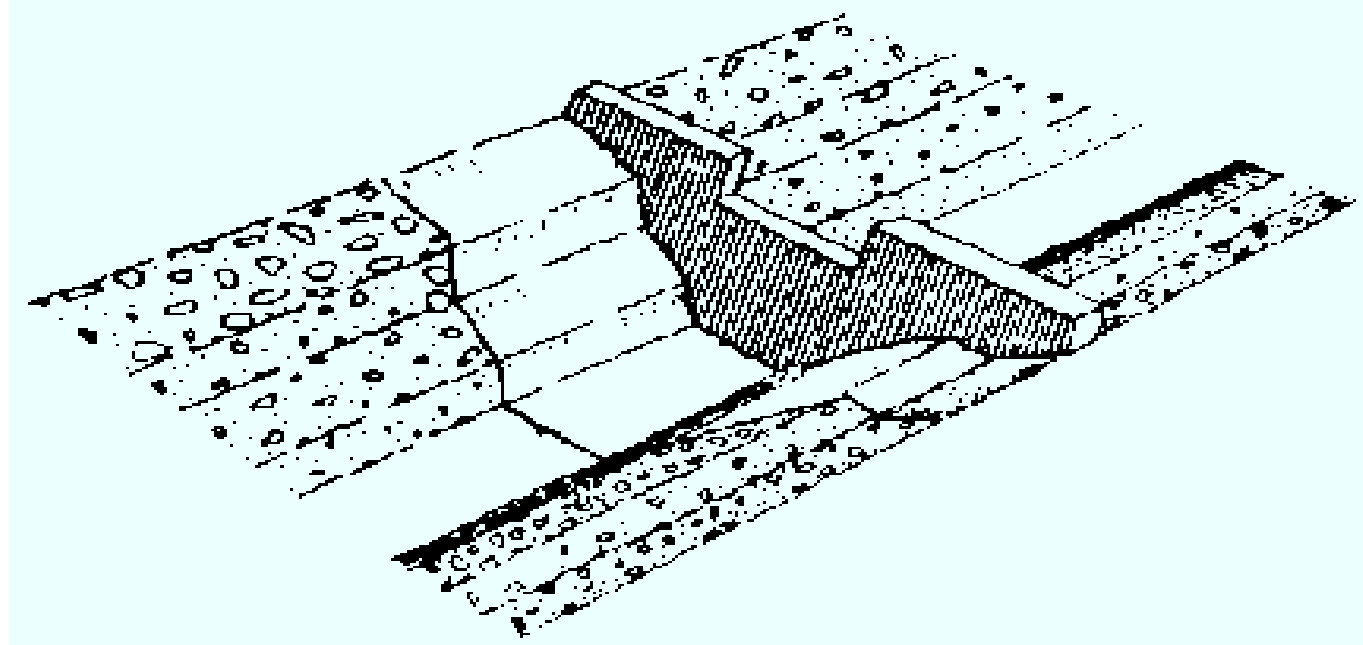


Figure: A permanent concrete check

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iii. Cross-Drainage Structures

- A Cross-drainage works or structure is a structure that carrying the canal across natural drainage, across other canals, across valleys or gulley and others obstacles.
- Are expensive, the alignment of the canal should be that w/c minimizes the cross-drainage works as much as possible.

Selection of types of cross drainage works

- However, in the actual field, ideal conditions may not be available and the choice would then depend upon the many factors, such as:
 - ☞ Relative bed level
 - ☞ Economical consideration
 - ☞ Discharge of the drainage
 - ☞ Construction problems
 - ☞ Suitable canal alignment

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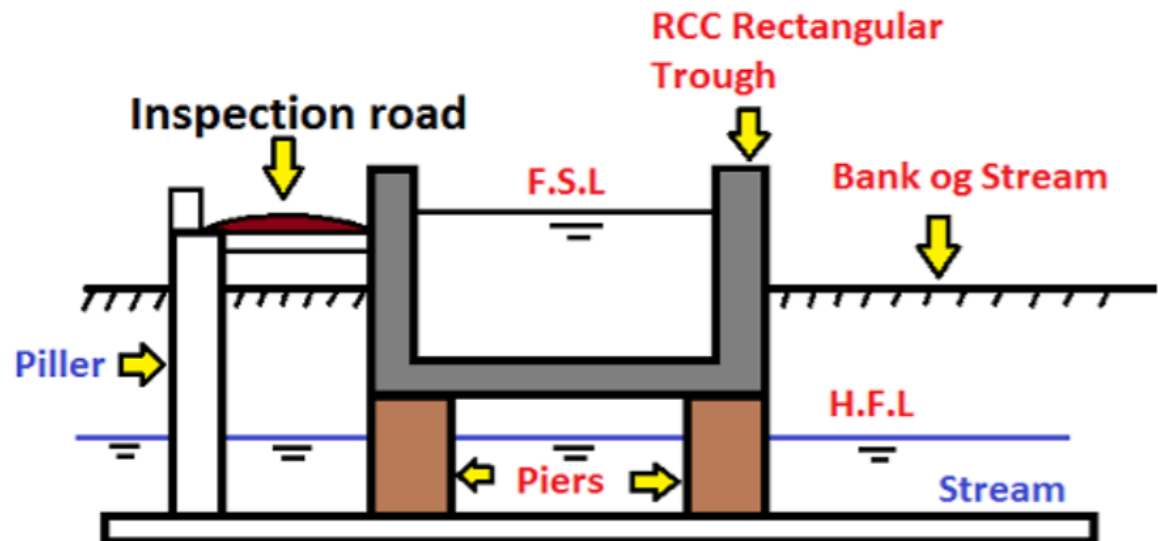
Types of Cross – Drainage Structures

- Based on the relative position of the canal and drainage, c-d works can be classified as:

1) Irrigation canal passing over the drainage:

a. Aqueduct

- When bed level of the canal is well above the HFL of drainage.



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Figure: Aqueduct

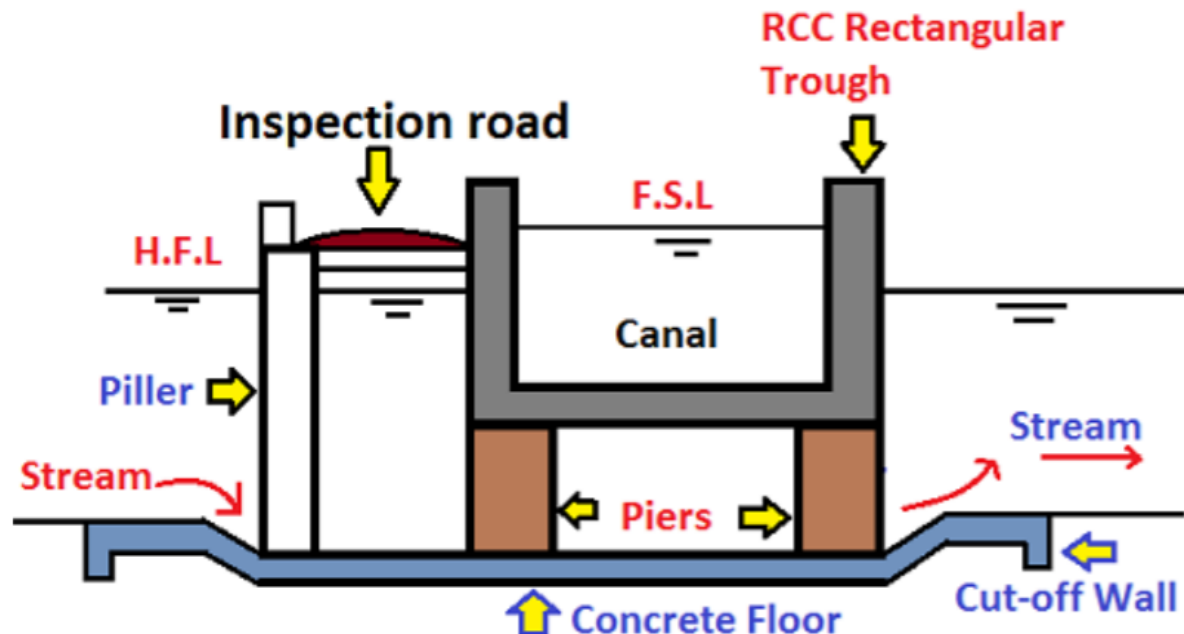
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Types of Cross – Drainage Structures . . .

1) Irrigation canal passing over the drainage . . .

b. Siphon aqueduct

- In the case of the siphon Aqueduct, the HFL (High Flow Level) of the drain is much higher above the canal bed, and
- Water runs under siphonic action through the aqueduct barrel.



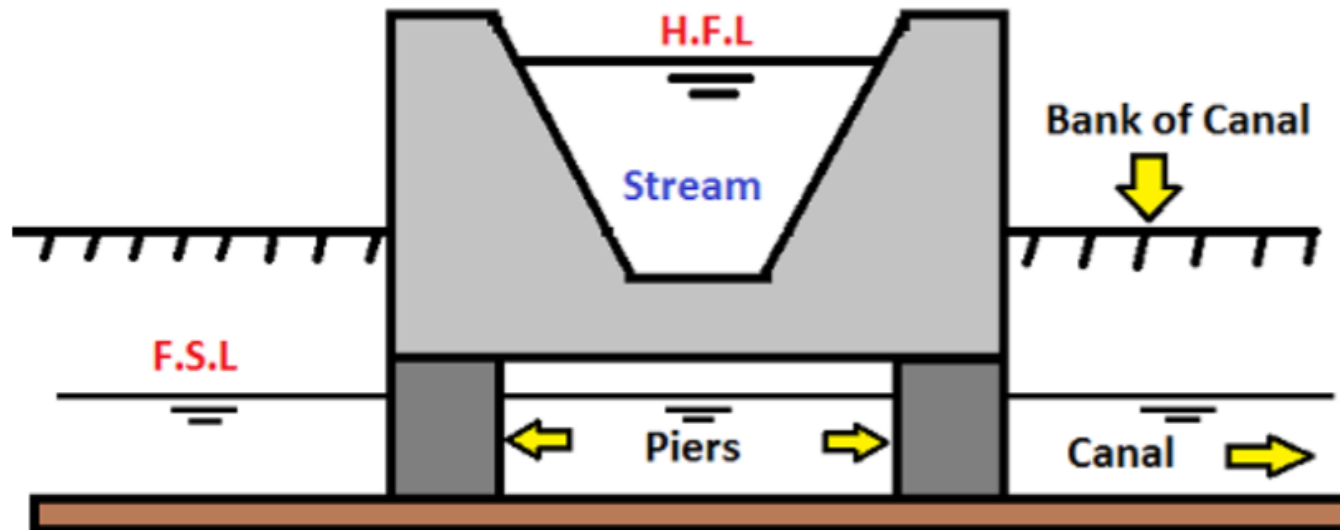
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Types of Cross – Drainage Structures . . .

2) Irrigation canal passing below the drainage:

a. Supper Passage

- This structure is suitable when the bed level of drainage is above the flood surface level of the canal.
- The water of the canal passes clearly below the drainage.



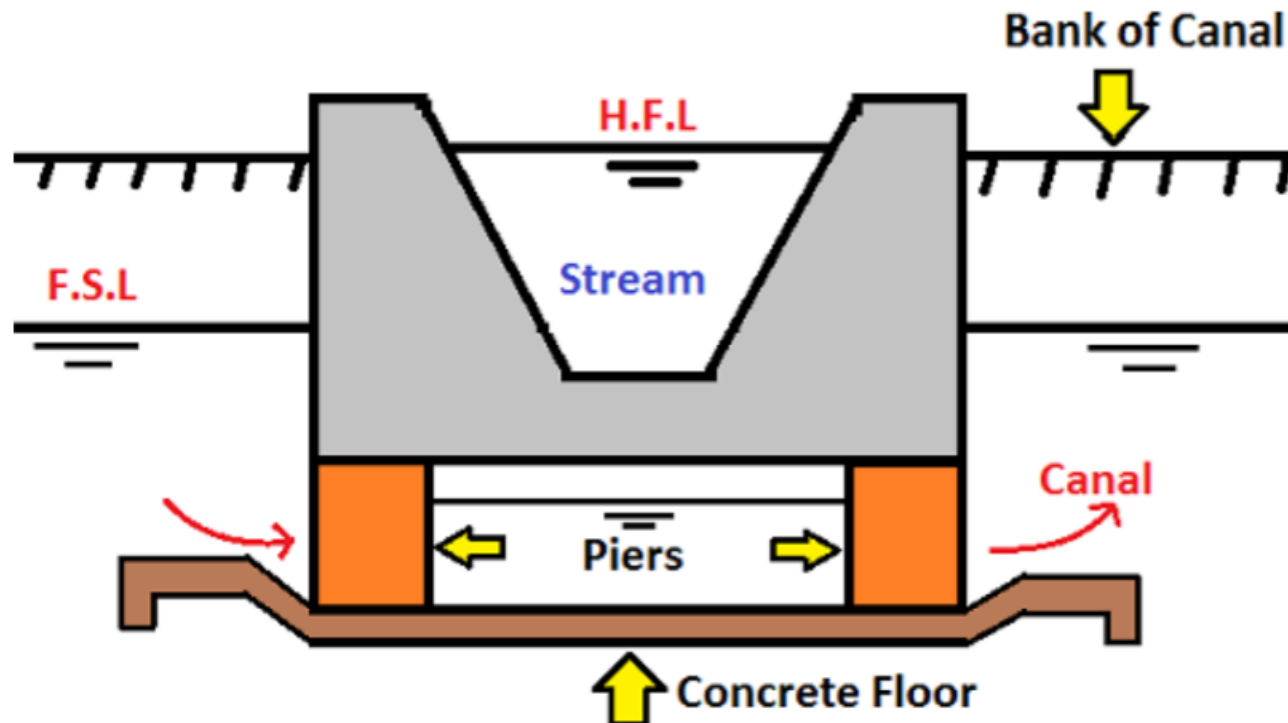
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Types of Cross – Drainage Structures . . .

2) Irrigation canal passing below the drainage:

b. Canal siphon

- The FSL of the canal is much above the bed level of the drainage trough, so that the canal runs under the siphonic action.



Cont'd...

❖ Based on functions, crossing structures can be classified as:

a. Flumes

- Flumes are used to carry irrigation water across gullies, ravines or other natural depressions.
- They are open canals made of wood (bamboo), metal or concrete which often need to be supported by pillars.

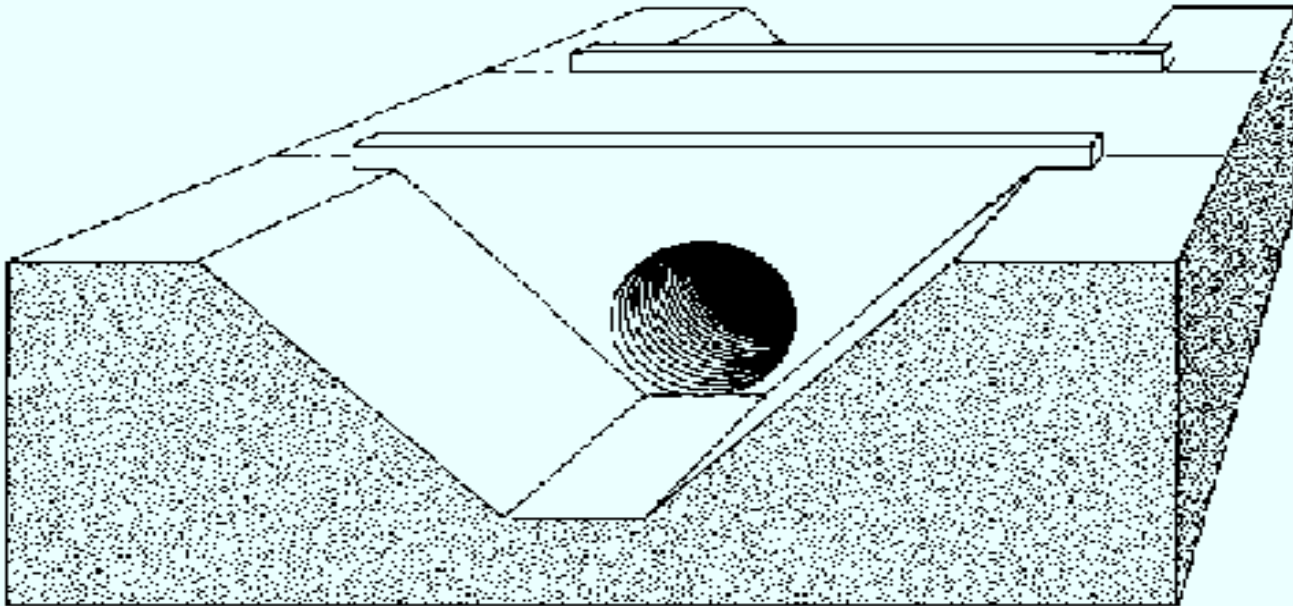


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❖ Based on functions, crossing structures can be classified as:

b. Culverts

- Culverts are used to carry the water across irrigation canals or roads.
- The structure consists of masonry or concrete headwalls at the inlet and outlet connected by a buried pipeline.



Cont'd...



Figure: Box Culverts